COMP396 DESIGN REPORT
COMP396 设计报告

PART 1: INDIVIDUAL TRADING STRATEGY DESIGNS
第一部分：个人交易策略设计

ASSET SELECTION PROCESS
资产选择过程

Data 1 includes 10 asset series (Series 01-10), and each series contains 1,000 trading days of OHLCV data. The preliminary visual analysis chart is as follows:
数据1包括10个资产系列（系列01-10），每个系列包含1000个交易日的 OHLCV 数据。初步的视觉分析图表如下：



Figure 0.1: Price evolution of all 10-asset series over the full 1000-day period
图0.1：全部10个资产系列在1000天期间的价格演变

Observed Characteristics:
观察到的特征：
Through visual inspection and preliminary analysis of 10 asset series, the following characteristics were determined:
通过对10个资产系列的目视检查和初步分析，确定了以下特征：

- **Series 01-03**: Mixed trending and ranging behavior with moderate volatility
- **系列01-03**：趋势与区间波动混合，波动性中等
- **Series 04**: Clear alternating trends with balanced upward/downward movements - suitable for trend-following strategies
- **04系列**：清晰的交替趋势，平衡的涨跌波动——适合趋势跟踪策略
- **Series 05**: Strong sustained uptrend with minimal drawdowns - ideal for buy-and-hold approach
- **05系列**：强劲的持续上升趋势，最低的回撤——非常适合买入并持有策略
- **Series 06-08**: High volatility with irregular patterns, challenging for systematic strategies
- **06-08系列**：波动性高且模式不规则，对系统性策略构成挑战
- **Series 09**: Range-bound oscillation around a stable mean - characteristic of mean-reverting behavior
- **09系列**：围绕稳定均值的区间振荡——均值回归行为的特征
- **Series 10**: Irregular price action with abrupt movements

Asset Selection Rationale:

- **第10系列**：不规则的价格走势和突然的波动**资产选**

择理由：

Based on the visual analysis, three assets were selected for strategy development:
基于视觉分析，选定三项资产用于战略制定：

- **Series 04**: Dual Moving Average Strategy
- **04系列**：双移动平均线策略
  - Reason: Series4 demonstrates clear trend phases, regular crossover opportunities and balanced directional movements, making it suitable for trend following based on MA.
  - 原因：Series4图表清晰呈现趋势阶段，具备规律的交叉机会和均衡的走势，因此适合基于移动平均线（MA）进行趋势跟踪。
- **Series 05**: Buy and Hold Strategy
- **05系列**：买入并持有策略
  - Reason: It shows a sustained long-term upward trend, has good risk-return characteristics, and the minimal reversion trend supports passive investment methods.
  - 原因：该指标呈现持续的长期上升趋势，具有良好的风险回报特性，且极小回归趋势支持被动投资策略。
- **Series 09**: Mean-Reversion Strategy
- **系列09**：均值回归策略
  - Reason: It shows stable range oscillations near the average price level, consistent reversion pattern suitable for statistical arbitrage.
  - 原因：该指数在均值附近呈现稳定的区间震荡，回归模式一致，适合进行统计套利。

The remaining seven assets exhibit characteristics (excessive volatility, irregular patterns or mixed signals) that make them less suitable for the type of target strategy.
其余七项资产表现出的特征（过度波动、不规则模式或混合信号）使其不太适合此类目标策略。

1.1 STRATEGY TYPE DEFINITION

1.1 战略类型定义

This trend-following strategy capitalizes on market momentum through golden cross and death cross signals from dual moving average crossovers. This method delivers optimal performance in clearly trending markets, excelling particularly during breakout phases that follow periods of consolidation. The strategy utilizes daily price charts with 3-day and 24-day moving averages, basing all trading decisions on daily closing prices while dynamically adapting position holding periods according to trend persistence. The approach builds on the well-documented momentum effect in financial markets - where assets showing strong recent performance tend to maintain their trajectory.

这种趋势追踪策略通过双移动平均线交叉产生的黄金交叉和死亡交叉信号，捕捉市场动能。该方法在趋势清晰的市场中表现最佳，尤其在震荡期后的突破阶段表现尤为突出。策略采用日线图结合3日和24日移动平均线，所有交易决策均基于当日收盘价，并根据趋势持续性动态调整持仓周期。该方法基于金融市场中广为人知的动量效应——近期表现强劲的资产往往能延续其上涨趋势。

1.2 DATA ANALYSIS

1.2 数据分析

Series 04 demonstrates clear trending characteristics, making it well-suited to dual moving average crossover strategies. The asset generates 65 crossover signals over a 1000-day trading period, yielding approximately 15 actionable signals annually with consistent trend persistence. Market direction shows a relatively balanced distribution, with 52.20% uptrend days and 47.80% downtrend days, lending support to a symmetric trading approach. The observed average trend duration of 15.2 days and a maximum duration of 59 days provide clear evidence of sustained directional moves, allowing MA crossover systems to effectively capture profitable trends.

04系列数据呈现出明显的趋势特征，非常适合采用双移动平均线交叉策略。该资产在1000个交易日中生成了65个交叉信号，每年可产生约15个可操作信号，且趋势持续性稳定。市场方向分布相对均衡，上涨日占52.20%，下跌日占47.80%，这为对称交易策略提供了有力支撑。观察到的平均趋势持续时间为15.2天，最长持续59天，充分证明趋势持续性，使得移动平均线交叉系统能有效捕捉有利趋势。

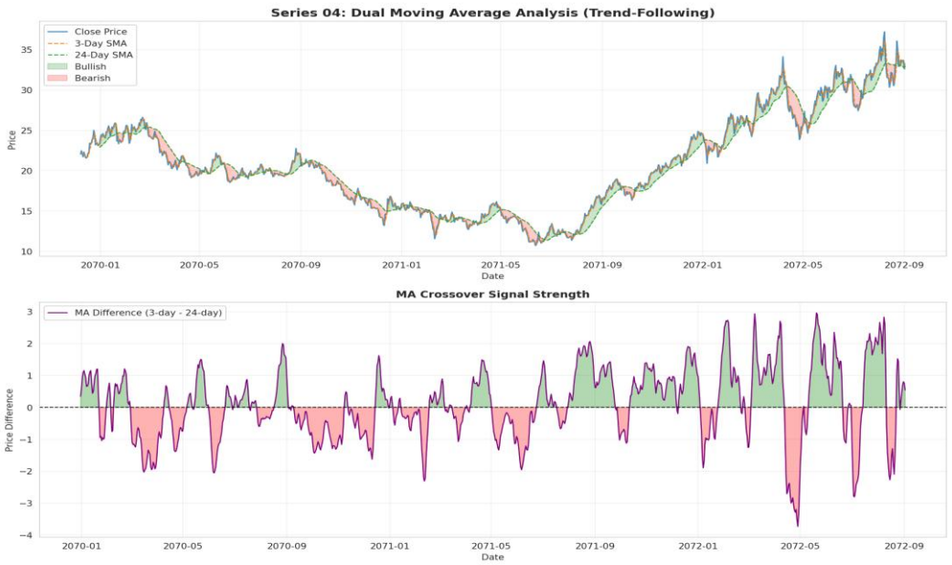


Figure 1.1: Series 04 price evolution with 3-day and 24-day SMAs showing trend identification

图1.1：04系列价格演变，3天和24天简单移动平均线显示趋势识别

The statistical analysis reveals an average signal persistence of 15.4 days per signal, indicating low whipsaw rates and meaningful trend development. The nearly 50-50 split between uptrends and downtrends allows the strategy to profit in both directions while reducing directional bias risk.

统计分析显示，每个信号的平均持续时间为15.4天，表明摆动率较低且趋势发展具有显著意义。由于上涨趋势与下跌趋势的占比接近50-50，该策略既能双向获利，又能有效降低方向性偏差风险。

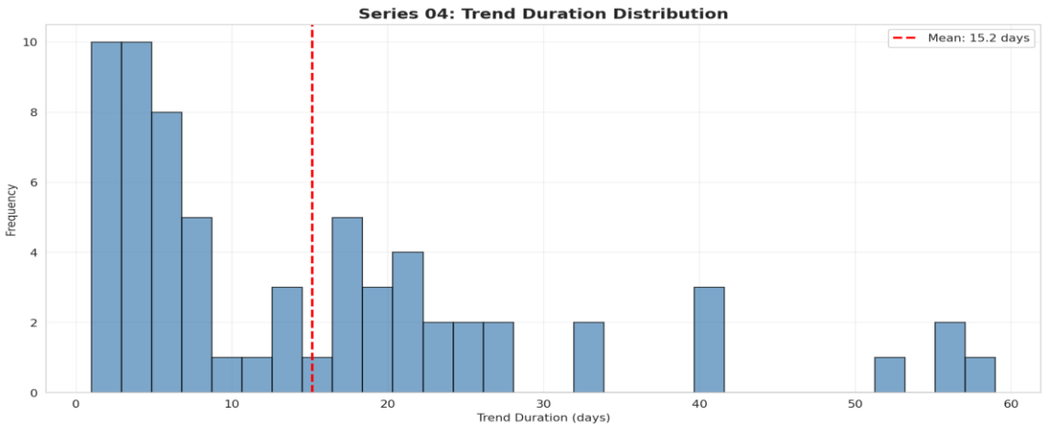


Figure 1.2: Distribution of trend durations in Series 04

图1.2：系列04中趋势持续时间的分布

1.3 STRATEGY TECHNICAL SPECIFICATION

1.3 战略技术规范

```
class DualMACrossover(bt.Strategy):
双重MACrossover策略（bt.）：
  params = (
  参数 = (
    ('fast_period', 3),      # Short-term MA period
      'fast_period' 3)，#短期移动平均周期
    ('slow_period', 24),     # Long-term MA period
      'slow_period' 24)，#长期移动平均周期
    ('stop_pct', 0.10),      # Stop-loss percentage
      'stop_pct' 0.10），#止损百分比
    ('target_pct', 0.20),    # Take-profit percentage
      'target_pct' 0.20），# 取利润百分比
    ('stake_pct', 0.25),     # Position size percentage
     "stake_pct" 0.25），# 仓位百分比
    ('series_index', 3),     # Series 04 (0-based index)
    （ 'series_index '3），#系列04（0作为索引）
  )
  )

  def __init__(self):
  def init（self）：
    self.fast_ma = bt.indicators.SMA(self.data.close, period=self.params.fast_period)
    self.fast_ma=bt.indicators.SMA（self.data. close，period=self. params.fast_period）
    self.slow_ma = bt.indicators.SMA(self.data.close, period=self.params.slow_period)
    self.slow_ma = bt.indicators.SMA（self.data. close，period=self. params.slow_period）
    self.crossover = bt.indicators.CrossOver(self.fast_ma, self.slow_ma)
    self.crossover = bt.indicators.CrossOver（self.fast_ma，self.slow_ma）

  def next(self):
  def next（self）：
    # Entry Logic: Golden Cross
    # 入场逻辑：黄金交叉
    if self.crossover > 0 and not self.position:
    如果自我.crossover >0且不自我.position：
      size = self.calculate_position_size()
      size =self.calculate_position_size（）
      self.buy(size=size)
      自我购买（size=size）
      self.entry_price = self.data.close[0]
      自我.entry_price=自我.data. close[0]

    # Exit Logic: Death Cross, Stop Loss, or Take Profit
    #退出逻辑：死亡十字、止损或止盈
    elif self.position:
    else self.position：
      current_price = self.data.close[0]
      当前价格=自己.数据.关闭[0]

      # Death cross exit
      # 死交叉退出
      if self.crossover < 0:
      如果自我<0交叉：
        self.sell(size=self.position.size)
        自我.卖（大小=自我.位置.大小）
    # Stop loss exit
    # 停损退出
    elif current_price <= self.entry_price * (1 - self.params.stop_pct):
    <sp_1elif current_price <=self.entry_price *（1-self. params.stop_pct）：
      self.sell(size=self.position.size)
      自我.卖（大小=自我.位置.大小）

    # Take profit exit
    # 实现盈利退出
    elif current_price >= self.entry_price * (1 + self.params.target_pct):
    如果当前价格大于或等于自己的1加上自己的params.target_pct：
      self.sell(size=self.position.size)
      自我.卖（大小=自我.位置.大小）
```

The strategy enters positions when the fast MA crosses above the slow MA (golden cross) with sufficient capital available and no existing position. Exits occur on death cross signals, 10% stop-loss, or 20% take-profit levels.

该策略在快速移动平均线（MA）向上穿越慢速移动平均线（MA）形成黄金交叉时入场，需满足资金充足且无持仓条件。止盈/止损触发条件包括：死亡交叉信号、10%止损位或20%止盈位。

1.4 PRELIMINARY TESTING AND OPTIMIZATION

1.4 初步测试与优化

DATA SPLITTING METHODOLOGY

数据分割方法

The Part 1 dataset (Days 1-1000) was divided using a 70/30 split for training and testing. The training phase, covering Days 1-700, focused exclusively on parameter optimization, while the testing period (Days 701-1000) provided out-of-sample data for final validation. This approach preserves the temporal sequence, treating the test set as future unseen data and delivering a realistic strategy assessment free from look-ahead bias.

第一部分数据集（第1-1000天）采用70/30的比例划分训练集和测试集。训练阶段（第1-700天）专注于参数优化，而测试阶段（第701-1000天）则提供样本外数据用于最终验证。该方法保留了时间序列特征，将测试集视为未来未见数据，从而实现无前瞻偏倚的策略评估。

PARAMETER OPTIMIZATION APPROACH

参数优化方法

The optimization employed a systematic grid search methodology testing multiple parameter combinations. Parameter ranges were selected based on technical analysis principles and the characteristics of dual moving average crossover systems:

该优化采用系统性网格搜索方法，对多种参数组合进行测试。参数范围的选择基于技术分析原则及双移动平均线交叉系统的特性：

- **Fast MA Period**: 3-15 days
- **快速MA周期：**3-15天
  - Rationale: Short-term momentum detection requires responsive moving averages.
  - 理论依据：短期动量检测需要响应迅速的移动平均线。
- **Slow MA Period**: 20-50 days
- **慢性MA期：**20-50天
  - Rationale: Medium-term trend identification across monthly cycles.
  - 依据：识别跨月周期的中期趋势。
- **Stop-Loss**: 5-15%
- **止损：**5-15%

  - Rationale: Balance between risk protection and avoiding premature exits.
  - 理论依据：在风险防护与避免过早退出之间寻求平衡。
- **Take-Profit**: 15-30%
- **止盈**：15-30%
  - Rationale: Capture significant trend moves while maintaining realistic profit targets.
  - 理论依据：在保持合理盈利目标的同时，捕捉重大趋势变动。
- **Position Size**: 20-30% portfolio value
- **仓位规模**：20-30%投资组合价值
  - Rationale: Meaningful market exposure while preserving capital for diversification.
  - 理论依据：在保留资本用于多元化投资的同时实现有效的市场敞口。

The grid search evaluated parameter combinations on the training dataset, employing the annualized Sharpe ratio as the primary optimization objective. This was supported by maximum drawdown and win rate metrics, which served as safeguards to ensure balanced performance characteristics.

网格搜索在训练数据集上评估参数组合，采用年化夏普比率作为主要优化目标。该方法通过最大回撤率和胜率指标提供支持，这些指标作为保障措施确保性能特征的平衡性。

OVERFITTING PREVENTION MEASURES

OVERFITTING 预防措施

To mitigate overfitting risk, a comprehensive validation approach need to be implemented. This included maintaining strict separation between the training (Days 1-700) and testing (Days 701-1000) periods, with the test set accessed only once for final validation to prevent data leakage. Parameter selection emphasized stability and robustness rather than maximizing any single metric, favoring combinations that demonstrated consistent performance. The optimization process favored simpler, standard configurations, which helps avoid overfitting to noise in the training data. Furthermore, the evaluation was comprehensive. Instead of focusing on just one metric, the system used a balanced assessment of the Sharpe ratio, maximum drawdown, and win rate together.

为降低过拟合风险，需要实施全面的验证方案。具体包括严格区分训练期（第1-700天）和测试期（第701-1000天），测试集仅在最终验证时使用一次以防止数据泄露。参数选择更注重稳定性和鲁棒性而非单一指标最大化，优先采用表现稳定的组合方案。优化过程倾向于采用更简洁的标准配置，有助于避免对训练数据噪声的过度拟合。此外，评估体系也做到全面均衡，系统不仅关注单一指标，还综合评估了夏普比率、最大回撤和胜率等指标。

PARAMETER SELECTION RATIONALE

参数选择依据

The final parameter set (fast=3, slow=24, stop=10%, target=20%, size=25%) was selected for its strong theoretical alignment with the strategy's design and demonstrated performance on training data. The 3-day fast setting enables responsive momentum detection, while the 24-day slow period (approximately one trading month) effectively captures medium-term trend development. The 10% stop-loss and 20% take-profit provide a 2:1 reward-risk ratio appropriate for trend-following strategies. The 25% position sizing enables meaningful exposure while maintaining portfolio diversification capacity.

最终确定的参数组合（快速=3天，慢速=24天，止损=10%，止盈=20%，仓位=25%）不仅与策略设计高度契合，还在训练数据中展现出优异表现。3天快速设置能实现即时动量捕捉，24天慢速周期（约相当于一个交易月）则能精准捕捉中期趋势走势。10%的止损率与20%的止盈率形成2：1的收益风险比，完美适配趋势跟踪策略。25%的仓位比例既保证了有效投资规模，又维持了投资组合的多元化配置能力。

FINAL PARAMETER CONFIGURATION

最终参数配置

Parameter 参数	Value 值	Justification 正当理由
Fast MA Period 快速MA周期	3 days 3天	Immediate momentum detection 瞬时动量检测
Slow MA Period 慢肌纤维期	24 days 24天	Monthly trend cycle identification 月度趋势循环识别
Stop-Loss 止损	10% 10%	Risk control with 2:1 reward-risk ratio 2：1风险报酬比下的风险控制
Take-Profit 获利了结	20% 20%	Target aligned with trendfollowing objectives 目标与趋势跟踪目标一致
Position Size 位置大小	25% 25%	Balance between exposure and diversification 风险敞口与分散化平衡

Metric 指标	Value 值
Test Return 测试返回	+17.37% +17.37%
Annualized Return 年化收益率	+14.45% +14.45%
Annualized Sharpe Ratio 年化夏普比率	2.013 2.013
Maximum Drawdown 最大回撤	2.58% 2.58%
Activity Level 活动水平	48.8% 48.8%
Win Rate 赢率	29.1% 29.1%
Trading Days 交易日	299 299

Table 1.1: Out-of-Sample Test Results (Days 701-1000)

表1.1：样本外试验结果（第701-1000天）

The strategy achieved exceptional out-of-sample performance with

该策略在样本外表现优异

annualized Sharpe ratio exceeding 2.0, demonstrating excellent risk-adjusted returns and robust parameter generalization. The moderate activity level of 48.8% indicates active but selective trading focused on high-quality trend signals.

年化夏普比率超过2.0，显示出卓越的风险调整回报率和稳健的参数泛化能力。48.8%的中等活跃度表明交易活动活跃但具有选择性，主要聚焦于高质量的趋势信号。

1.5 PLAN

1.5 计划

As a trend-following approach, Strategy S1 (Dual MA Crossover) demonstrates its established limitation in Data 1 through weak performance during range-bound "whipsaw" conditions. To address this vulnerability, the Semester 2 development plan will incorporate a market regime filter. Our investigation will focus on the Average Directional Index (ADX), testing the hypothesis that restricting new buy signals to ADX values above a specific threshold (e.g., 20-25) can effectively filter out low-probability signals during consolidation periods. We will first assess the filter's preliminary efficacy using the complete Data 1 history. For robust adaptation to current market conditions, primary development will then transition to Data 2, where we will partition the data into new training and out-of-sample test sets (e.g., 70/30 split). This framework will enable ADX threshold recalibration and final validation in the new market environment.

作为一种趋势跟踪方法，策略S1（双移动平均线交叉）在数据1中通过区间波动“鞭锯”条件下的表现不佳，暴露了其固有的局限性。为了应对这一弱点，第二学期的发展计划将引入市场机制过滤器。我们的研究将重点关注平均方向指数（ADX），测试假设，即仅在 ADX 值超过特定阈值（例如20-25）时发出新的买入信号，可以有效过滤掉盘整期间低概率的信号。我们将首先使用完整的数据1历史记录评估该过滤器的初步效能。为了更好地适应当前市场条件，主要开发工作随后将转向数据2，在此我们将数据划分为新的训练集和样本外测试集（例如70/30分割）。这一框架将使 ADX 阈值的重新校准和最终验证在新的市场环境中得以实现。

STRATEGY 2: BUY AND HOLD (SERIES 05)

策略2：买入并持有（系列05）

2.1 STRATEGY TYPE DEFINITION

2.1 战略类型定义

This strategy employs a passive trend-following methodology specifically optimized for assets demonstrating sustained upward trends. The approach performs particularly effectively during sustained uptrends with consistent momentum, achieving optimal results in low volatility environments relative to trend magnitude. Operating as a pure buy-and-hold approach with a single entry point at initiation and no subsequent trading decisions, the strategy builds upon the theoretical foundations of the efficient market hypothesis and the long-term equity premium principle.

该策略采用被动趋势跟踪方法，专门针对呈现持续上涨趋势的资产进行优化。在具有稳定动能的持续上涨行情中，该策略表现尤为出色，尤其在趋势幅度与波动性较低的市场环境中能取得最佳效果。作为纯粹的买入并持有策略，该方案仅在初始阶段设定单一入场点，后续不再进行交易决策。其理论基础源于有效市场假说和长期股票溢价原则。

2.2 DATA ANALYSIS

2.2 数据分析

Series 05 exhibits pronounced uptrend characteristics that strongly align with a buy-and-hold investment strategy. Over the 1000-day trading period, the asset delivered a total return of 34.53%, equating to an annualized return of 7.76%. Its statistical profile reveals that over half of all trading days (54.63%) closed positively, while the price traded above its 50-day moving average for 63.10% of the period—both indicators confirming sustained upward momentum. With a maximum drawdown limited to 10. 13%, the strategy achieved a favorable return-to-drawdown ratio of 3.4:1, reflecting strong risk-adjusted performance.

05系列指数展现出显著的上升趋势特征，与买入并持有投资策略高度契合。在1000个交易日中，该资产实现总回报率34.53%，折合成年化回报率为7.76%。统计数据显示，超过半数交易日（54.63%）收涨，且有63.10%的交易日价格高于50日移动平均线——这两个指标均印证了持续的上行动能。该策略最大回撤率控制在10.13%，实现了3.4：1的优异回撤率，充分体现了经过风险调整后的强劲表现。

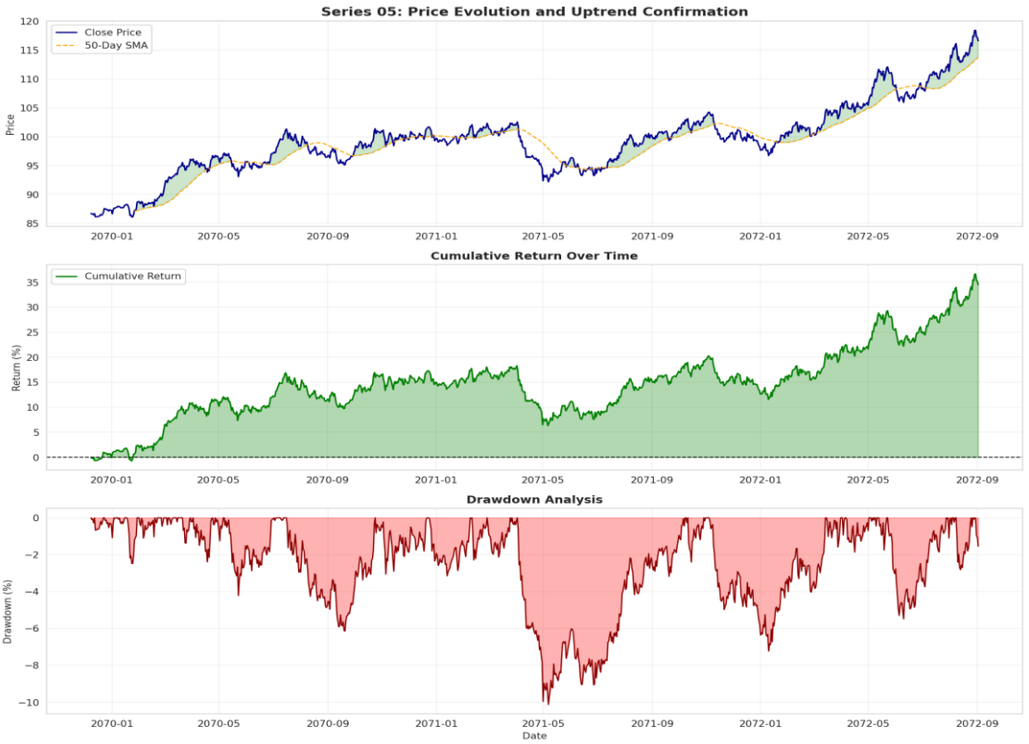


Figure 2.1: Series 05 comprehensive uptrend analysis showing sustained growth

图2.1：05系列全面上升趋势分析显示持续增长

Daily returns exhibit a positive mean of 0.031% with volatility of 0.505%, creating an ideal environment for passive strategies where positive drift compounds with minimal noise.

日收益率呈现0.031%的正均值，波动率为0.505%，为被动策略创造了理想环境，其中正向漂移与最小噪声相叠加。

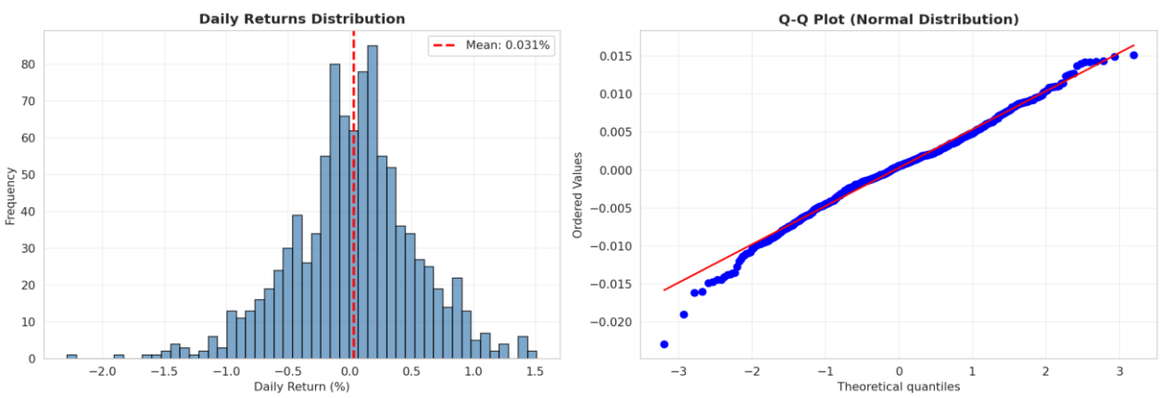


Figure 2.2: Series 05 returns distribution showing positive drift

图2.2：系列05返回分布显示正漂移

2.3 TECHNICAL SPECIFICATION

2.3 技术规格

```
class BuyAndHold(bt.Strategy):
BuyAndHold策略（bt. Strategy）：
  params = (
  参数 = (
    ('series_index', 4),    # Series 05 (0-based index)
    （‘series_index‘4），# 系列 05（基于 0 的索引）
    ('stake', 100),        # Deploy all available capital
     “股权”100），#部署所有可用资本
  )
  )

  def __init__(self):
  def init（self）：
    self.bought = False
    自己。购买=假货

  def next(self):
  def next（self）：
    # Buy maximum shares on first trading day
    # 交易首日最大持股量
    if not self.bought and not self.position:
    如果不是自我购买，那么自我的立场是：
      available_cash = self.broker.get_cash()
      available_cash=self. broker.get_cash（）；
      current_price = self.data.close[0]
      当前价格=自己.数据.关闭[0]
      max_shares = int(available_cash / current_price)
      max_shares =int（可用现金/当前价格）

      if max_shares > 0:
      如果max_shares >0：
        self.buy(size=max_shares)
        自行购买（规模=最大份额）
        self.bought = True
        自己。购买=真实
```

The strategy executes a single entry on the first trading day deploying maximum available capital, then holds the position until strategy termination with no intermediate exit decisions.

该策略在首个交易日执行单笔入场操作，投入最大可用资本，随后持有该头寸直至策略终止，期间不进行任何中途退出决策。

2.4 PRELIMINARY TESTING AND SERIES SELECTION

2.4 初步测试与系列选择

ASSET SELECTION METHODOLOGY

资产选择方法

Unlike parametric strategies that demand extensive parameter optimization, the buy-and-hold approach centers on careful asset selection. The core optimization task lies in identifying which of the ten available series exhibits the most favorable uptrend characteristics for passive investment—a methodical selection process that utilized the training period (Days 1-700) for initial assessment.

与需要大量参数优化的参数化策略不同，买入并持有策略的核心在于精挑细选资产。其核心优化任务在于从十个可选系列中筛选出最适合被动投资的上行趋势特征——这一系统性筛选过程利用了训练期（第1-700天）进行初步评估。

SERIES SELECTION PROCESS

系列选择过程

The training period assessment evaluated multiple series across essential metrics—including total return, risk-adjusted performance (Sharpe ratio), maximum drawdown, percentage of positive days, and time spent above key moving averages. Among these, Series 05 stood out as the preferred candidate, displaying robust uptrend characteristics coupled with favorable risk-adjusted returns. Its consistent positive momentum and controlled drawdown levels further validated its suitability for a passive buy-and-hold strategy. The selection was based on multiple convergent metrics rather than a single performance measure, reducing the risk of selecting an asset that performed well due to random variation. The analysis focused on persistent structural characteristics such as trend consistency and drawdown profiles rather than short-term patterns.

训练期评估通过综合多项关键指标对多个系列进行分析，包括总回报率、风险调整收益率（夏普比率）、最大回撤幅度、正收益天数占比以及突破关键移动平均线的时间。其中，05系列脱颖而出成为首选，不仅展现出强劲的上升趋势，还具备良好的风险调整收益。其持续的正向动能和可控的回撤水平进一步印证了其适合被动持有策略。该选择基于多重趋同指标而非单一表现指标，有效降低了因随机波动而选中表现优异资产的风险。分析重点聚焦于趋势持续性和回撤特征等结构性特征，而非短期波动模式。

CAPITAL DEPLOYMENT DECISION

资本配置决策

For buy-and-hold strategies, capital allocation represents the fundamental strategic decision. Given the scale-invariant nature of the Sharpe ratio in single-asset, non-leveraged implementations, deploying 100% of capital was identified as the optimal approach to maximize absolute returns without compromising the original risk-adjusted performance characteristics. The intentional avoidance of leverage further guarantees that risk exposure remains naturally constrained by the asset's inherent volatility profile.

在买入并持有策略中，资本配置是核心战略决策。鉴于单资产非杠杆化操作中夏普比率的规模不变特性，将100%资本投入被确定为最优方案——既能实现绝对收益最大化，又不损害风险调整后的原始绩效特征。刻意规避杠杆操作，进一步确保风险敞口始终受制于资产固有的波动性特征。

OVERFITTING CONSIDERATIONS

OVERFITTING 考虑事项

Buy-and-hold strategies inherently present minimal overfitting risk compared to parametric approaches, as their optimization focuses on asset selection grounded in fundamental trend characteristics rather than parameter fitting susceptible to training data noise. The selection methodology prioritized identifying persistent structural features observable throughout the training period, rather than optimizing specific numerical parameters.

与参数化方法相比，买入并持有策略本质上具有更低的过拟合风险，因为其优化重点在于基于基本面趋势特征的资产选择，而非易受训练数据噪声影响的参数拟合。该选择方法优先识别贯穿整个训练周期的持续性结构性特征，而非优化特定数值参数。

Metric 指标	Value 值
Test Return 测试返回	+14.31% +14.31%
Annualized Return 年化收益率	+11.93% +11.93%

Metric 指标	Value 值
Annualized Sharpe Ratio 年化夏普比率	1.335 1.335
Maximum Drawdown 最大回撤	5.82% 5.82%
Activity Level 活动水平	99.0% 99.0%
Win Rate 赢率	54.5% 54.5%
Trading Days 交易日	299 299

Table 2.1: Out-of-Sample Performance (Days 701-1000)

表2.1：样本外性能（第701-1000天）

The out-of-sample results confirm the strategy's effectiveness, showing robust performance with an annualized Sharpe ratio of 1.335 and a 14.31% return. Throughout the testing period, the strategy consistently exhibited its core uptrend-following behavior without displaying overfitting tendencies. The 99.0% activity level reflects continuous market exposure, fully aligning with the buyand-hold operational mandate.

样本外测试结果印证了该策略的有效性，其稳健表现体现在年化夏普比率1.335和14.31%的收益率上。在测试期间，该策略始终遵循核心的上行趋势跟踪行为，未出现过拟合倾向。99.0%的活跃度水平表明持续的市场敞口，完全符合买入并持有操作策略的要求。

2.5 PLAN

2.5 计划

The principal limitation of Strategy S2 (Buy and Hold) lies in its absence of active risk management during trades, resulting in full exposure during market declines. The Semester 2 development plan aims to transform this passive approach into a dynamic macro strategy through implementation of a regime-based filter. We will specifically investigate the 200-day moving average (200-DMA) as a potential mechanism, testing the rule: "maintain positions only when the asset's closing price remains above its 200-DMA." This modification is designed to enable systematic position exits and capital preservation during significant bear markets—conditions we anticipate encountering in Data 2. Preliminary backtests on Data 1 will assess conceptual viability, while primary validation will utilize Data 2 partitioned into dedicated training and test sets to definitively evaluate the filter's effectiveness in mitigating major drawdowns within the new market environment.

策略S2（买入并持有）的主要局限在于交易过程中缺乏主动风险管理，导致市场下跌时完全暴露风险。第二学期的发展计划旨在通过实施基于制度的筛选机制，将这种被动策略转变为动态宏观策略。我们将重点研究200日移动平均线（200-DMA）作为潜在机制，验证“仅在资产收盘价保持在200-DMA上方时维持头寸”的规则。这一调整旨在使系统化平仓和资本保值成为可能——我们预计在数据2中会遇到显著的熊市行情。对数据1的初步回测将评估概念可行性，而主要验证将利用数据2划分为专用训练集和测试集，以明确评估该筛选机制在新市场环境下缓解重大回撤的有效性。

STRATEGY 3: STATISTICAL MEAN-REVERSION (SERIES 09)

策略3：统计均值回归（系列09）

3.1 STRATEGY TYPE DEFINITION

3.1 战略类型定义

This strategy employs a statistically grounded mean-reversion approach using Z-score methodology to identify significant price deviations. Designed for range-bound markets displaying oscillatory price action, the strategy capitalizes on the tendency of prices to revert toward a statistical mean. The methodology operates with short-to-medium holding periods—typically 5 to 20 days—and relies on the well-documented mean-reversion phenomenon observed in financial markets, where prices reaching statistical extremes relative to historical patterns tend to undergo subsequent corrections.

该策略采用基于统计学的均值回归方法，运用Z分数分析法识别显著价格偏离。专为呈现震荡行情的区间市场设计，策略充分利用价格回归统计均值的特性。其操作周期为中短期（通常5至20天），基于金融市场中被充分验证的均值回归现象——当价格偏离历史走势达到统计学极端值时，往往会出现后续回调。

3.2 DATA ANALYSIS

3.2 数据分析

Series 09 demonstrates strong mean-reverting properties ideal for statistical arbitrage approaches. The asset generated 39 oversold signals (Z-score < -2) and 69 overbought signals (Z-score > +2), yielding approximately 11 annual trading opportunities. The mean-reversion correlation of -0.049 between current Z-score and subsequent 5-day returns confirms the statistically significant tendency for price extremes to reverse. Furthermore, with prices remaining within one standard deviation 41.71% of the time, the series exhibits the range-bound conditions essential for effective mean-reversion strategies.

09号序列展现出优异的均值回归特性，堪称统计套利策略的理想标的。该资产生成了39个超卖信号（Z值< -2）和69个超买信号（Z值> +2），共产生约11个年度交易机会。当前Z值与后续5日收益率之间的均值回归相关系数为-0.049，证实了价格极端值反转的统计显著性趋势。此外，由于41.71%的时间内价格波动范围控制在单个标准差内，该序列完全符合有效均值回归策略所需的区间波动条件。

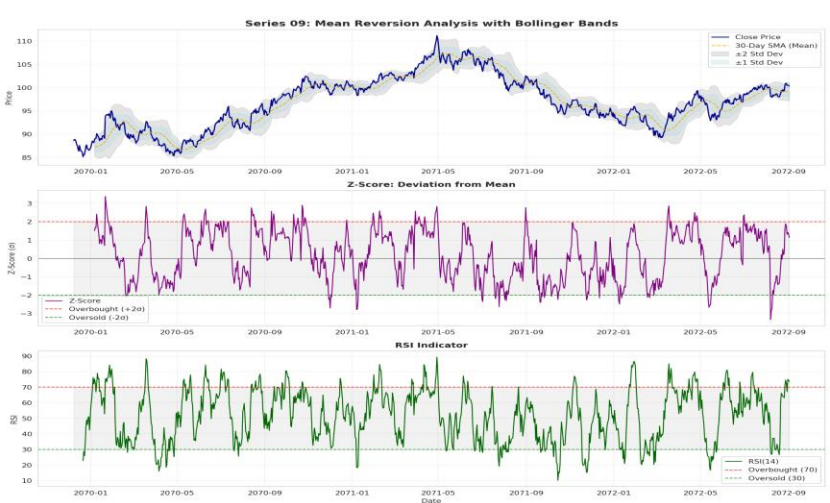


Figure 3.1: Series 09 mean-reversion indicators with Bollinger Bands and Z-score

图3.1：09系列均值回归指标与布林带和Z值

The Z-score distribution shows a near-zero mean of 0.179 with moderate standard deviation of 1.338, indicating price fluctuations centered around the moving average.

Z分数分布显示均值接近零（0.179），标准差为1.338，表明价格波动主要集中在移动平均线附近。

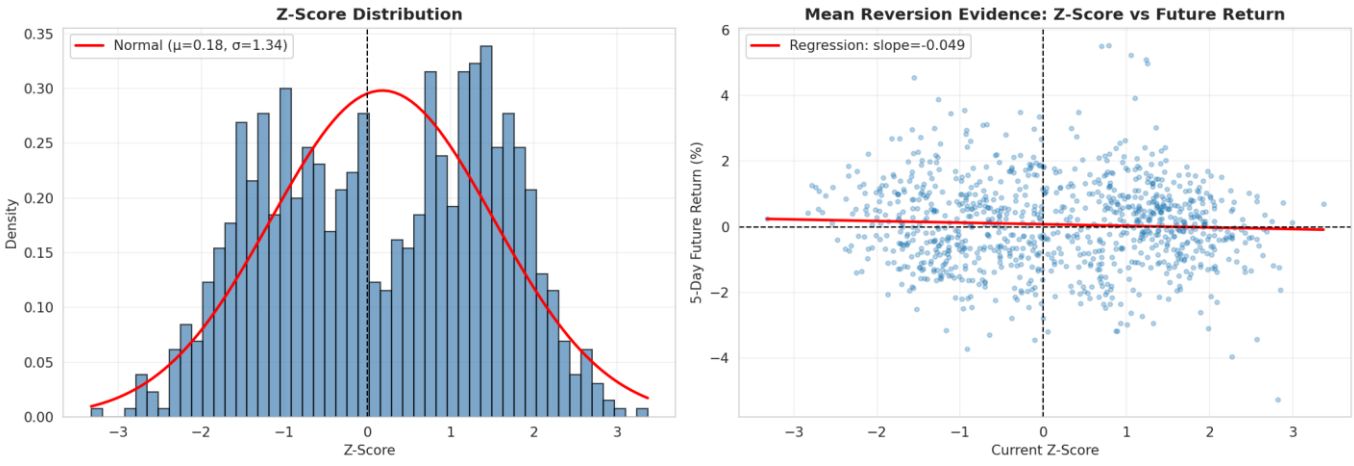


Figure 3.2: Z-score distribution and mean-reversion evidence

图3.2：Z值分布和均值回归证据

3.3 TECHNICAL SPECIFICATION

3.3 技术规格

```
class MeanReversionStrategy(bt.Strategy):
class MeanReversionStrategy（bt. 策略）：
  params = (
  参数 = (
    ('mean_period', 30),      # Mean calculation period
      'mean_period' 30)，#平均计算周期
    ('entry_threshold', 2.0),  # Z-score entry threshold
      'entry_threshold' 2.0），#Z分数输入阈值
    ('exit_threshold', 0.5),   # Z-score exit threshold
      'exit_threshold' 0.5），#Z分数退出阈值
    ('rsi_period', 14),       # RSI confirmation period
      'rsi_period' 14)，#RSI确认周期
    ('max_holding_days', 20),  # Maximum position duration
      'max_holding_days' 20），# 最大位置持续时间
    ('stake_pct', 0.15),      # Position size percentage
     "stake_pct" 0.15），# 仓位百分比
    ('series_index', 8),      # Series 09 (0-based index)
    （'series_index' 8），# 系列 09（基于 0 的索引）
  )
  )
  def __init__(self):
  def init（self）：
    self.mean_price = bt.indicators.SMA(self.data.close, period=self.params.mean_period)
    self.mean_price = bt.indicators.SMA（self.data. close，period=self. params.mean_period）
    self.std_dev = bt.indicators.StdDev(self.data.close, period=self.params.mean_period)
    self.std_dev = bt.indicators.StdDev（self.data. close，period=self. params.mean_period）
    self.rsi = bt.indicators.RSI(self.data.close, period=self.params.rsi_period)
    self.rsi = bt.indicators.RSI（self.data. close，period=self. params.rsi_period）

  def next(self):
  def next（self）：
    z_score = (self.data.close[0] - self.mean_price[0]) / self.std_dev[0]
    z_score =（self.data. close[0]-self.mean_price[0]）/self.std_dev[0]

    # Entry: Z-score <= -2.0 and RSI <= 30
    # 入场条件：Z值 <= -2.0且相对强弱指数 <= 30
    if not self.position and z_score <= -2.0 and self.rsi[0] <= 30:
      size = self.calculate_position_size()
      size =self.calculate_position_size（）
      self.buy(size=size)
      自我购买（size=size）

    # Exit: Z-score >= 0.5 or risk management triggers
    # 终止：Z值≥0.5或触发风险管理机制
    elif self.position:
    else self.position:
      if z_score >= 0.5 or self.check_risk_exits():
        self.sell(size=self.position.size)
        自我.卖（大小=自我.位置.大小）
```

The strategy enters when Z-score falls below -2.0 with RSI confirmation below 30 and exits when Z-score rises above 0.5 or risk management trigger activates.

该策略在Z值低于-2.0且相对强弱指数（RSI）确认值低于30时入场，当Z值升至0.5以上或风险管理触发条件激活时退出。

3.4 PRELIMINARY TESTING AND OPTIMIZATION

3.4 初步测试与优化

DATA SPLITTING AND OPTIMIZATION FRAMEWORK

数据分割与优化框架

Following the same train-test methodology as Strategy 1, the optimization process used Days 1-700 for parameter tuning and Days 701-1000 for out-of-sample validation. Mean-reversion strategies require careful parameter calibration to balance signal sensitivity against false positives in ranging markets.

采用与策略1相同的训练-测试方法，优化过程使用第1-700天进行参数调校，第701-1000天进行样本外验证。均值回归策略需要谨慎校准参数，以平衡市场波动中的信号灵敏度与假阳性率。

PARAMETER OPTIMIZATION APPROACH
参数优化方法

The optimization explored a multidimensional parameter space through systematic grid search testing multiple combinations:
该优化研究通过系统性网格搜索测试多种组合，探索了多维参数空间：

- **Mean/Standard deviation Period**: 20-40 days
- **平均值/标准差周期：**20-40天
  - Rationale: Statistical windows must capture meaningful price distribution while remaining responsive.
  - 理论依据：统计学窗口必须在保持灵敏度的同时，能够捕捉到具有统计学意义的价格分布。
- **Entry Threshold (Z-score)**: 1.5-2.5
- **进入阈值（Z值）：**1.5-2.5
  - Rationale: Balance between signal frequency and statistical significance.
  - 理论依据：信号频率与统计学显著性之间的平衡。
- **Exit Threshold (Z-score)**: 0.0-1.0
- **退出阈值（Z值）：**0.0-1.0
  - Rationale: Capture mean-reversion move while avoiding premature exits.
  - 理论依据：在避免提前退出的同时，捕捉均值回归移动。
- **RSI Period**: 10-20 days
- **RSI周期：**10-20天
  - Rationale: Confirmation indicator for oversold/overbought conditions.
  - 理论依据：超卖/超买状态的确认指标。
- **Maximum Holding Days**: 10-30 days
- **最长持有天数：**10-30天
  - Rationale: Prevent capital lock-up in failed reversions.
  - 依据：防止资本在失败的反向收购中被锁定。
- **Position Size**: 10-20%
- **位置大小：**10-20%
  - Rationale: Conservative sizing appropriate for contrarian strategies.
  - 理论依据：保守的尺寸选择适用于逆向策略。

The primary optimization objective was Sharpe ratio, supplemented by evaluation of win rate and maximum drawdown constraints to ensure the strategy maintained acceptable risk characteristics and avoided excessive losses from failed mean-reversion signals.
主要优化目标为夏普比率，同时辅以胜率评估和最大回撤约束，以确保策略保持可接受的风险特征，并避免因均值回归信号失效导致的过度亏损。

OVERFITTING PREVENTION MEASURES
OVERFITTING 预防措施

Mean-reversion strategies face substantial overfitting risk since historical ranging patterns may not continue in future periods. To address this, multiple systematic safeguards need to be implemented: Use the minimum trade count filter exclude parameter combinations generating insufficient trading activity during training, as small sample sizes prevent reliable performance assessment; Multi-period validation tested parameter stability by examining performance across various market environments within the training data, ensuring robustness under different conditions rather than optimizing for any single favorable regime; Drawdown constraints rejected combinations exceeding conservative maximum drawdown thresholds on training data, prioritizing capital preservation in line with mean-reversion principles; Strict out-of-sample testing completely isolate the testing period (Days 701-1000) from optimization, with this data accessed only once for final validation.
均值回归策略存在显著的过拟合风险，因为历史波动模式可能无法在未来持续。为解决这一问题，需要实施多项系统性保障措施：使用最小交易量筛选器排除训练期间交易活动不足的参数组合，因样本量过小导致性能评估不可靠；通过多周期验证测试参数稳定性，通过分析训练数据中不同市场环境下的表现，确保模型在不同条件下的稳健性而非针对单一有利市场进行优化；采用回撤约束排除训练数据中超出保守最大回撤阈值的组合，优先遵循均值回归原则保护资本；严格进行样本外测试，将测试期（第701-1000天）与优化过程完全隔离，仅允许最终验证时访问一次该数据。

PARAMETER SELECTION RATIONALE
参数选择依据

The finalized parameters (entry Z=-2.0, exit Z=+0.5, mean period=30, RSI=14, max holding=20, size=15%) were selected based on their theoretical coherence and demonstrated performance during training. The entry threshold of -2.0 standard deviations represents a conservative statistical extreme—approximately the 2.3% probability level under normal distribution—effectively balancing the reduction of false signals against the identification of genuinely oversold conditions. Similarly, the exit threshold at +0.5 standard deviations captures meaningful partial mean-reversion without requiring a full return to the mean, which may not materialize within practical time horizons. The 30-day statistical window provides an optimal compromise between responsiveness and distribution stability, while the 14-day RSI period offers independent momentum confirmation. Finally, the 20-day maximum holding period enforces disciplined risk management by preventing indefinite capital commitment to positions that fail to revert as anticipated.
最终确定的参数设置（入场Z=-2.0，离场Z=+0.5，平均周期30天，相对强弱指数14，最大持有20天，仓位比例15%）基于理论一致性及训练阶段的实证表现。-2.0标准差的入场阈值采用保守统计极限——在正态分布下对应约2.3%的概率水平——有效平衡了误判信号的减少与真实超卖状态的识别。+0.5标准差的离场阈值则捕捉到有意义的部分均值回归，无需完全回归均值，这在实际操作中可能难以实现。30天统计窗口在响应速度与分布稳定性之间取得最佳平衡，14天相对强弱指数周期则提供独立的动量验证。最后，20天的最大持有期通过防止对未按预期回归的仓位进行无限期资金投入，强化了纪律化的风险管理。

FINAL PARAMETER CONFIGURATION
最终参数配置

Parameter 参数	Value 值	Justification 正当理由
Entry Threshold (Z-score) 进入阈值（Z值）	-2.0 -2.0	Statistical extreme (2.3% probability) 统计学极端值（2.3%概率）
Exit Threshold (Z-score) 退出阈值（Z值）	+0.5 +0.5	Captures partial reversion, avoids greed 捕获部分逆转，避免贪婪
Mean Period 平均周期	30 days 30天	Monthly statistical window 月度统计窗口
RSI Period RSI周期	14 days 14天	Standard momentum confirmation 标准动量确认
RSI Oversold 相对强弱指数超卖	30 30	Conservative oversold threshold 保守超卖阈值
Maximum Holding 最高持股	20 days 20天	Limits capital lock-up risk 限制资本锁定期风险
Position Size 位置大小	15% 15%	Conservative contrarian sizing 保守反向定额
Stop-Loss 止损	5% 5%	Tight risk control for failed reversions 对失败逆转的严格风险控制

Metric 指标	Value 值
Test Return 测试返回	+0.27% +0.27%
Annualized Return 年化收益率	+0.23% +0.23%
Annualized Sharpe Ratio 年化夏普比率	0.396 0.396
Maximum Drawdown 最大回撤	0.37% 0.37%
Activity Level 活动水平	12.7% 12.7%
Win Rate 赢率	6.7% 6.7%
Trading Days 交易日	300 300

Table 3.1: Out-of-Sample Performance (Days 701-1000)
表3.1：样本外性能（第701-1000天）

Parameter 参数	Value 值	Justification 正当理由
Take-Profit 获利了结	4% 4%	Modest target aligned with reversion magnitude 与回复幅度相一致的适度目标

The out-of-sample results confirm sustained positive risk-adjusted performance with a Sharpe ratio of 0.396, validating the parameter selection framework. While absolute returns remain modest at 0.27%, the strategy demonstrates exceptional drawdown control (0.37%) while maintaining positive performance despite challenging market conditions. The 12.7% activity rate underscores the strategy's selective nature, adhering strictly to its conservative Z-score entry requirements. The significant win rate decline—from 30% during training to just 6.7% in testing—indicates substantially weaker mean-reversion patterns during the evaluation period. This performance divergence highlights why conservative parameters and disciplined risk protocols remain essential for strategy durability.

样本外测试结果印证了该策略持续保持正向风险调整收益，夏普比率达0.396，验证了参数选择框架的有效性。尽管绝对回报率仅为0.27%，但该策略在市场环境严峻时仍保持正收益，同时展现出卓越的回撤控制能力（0.37%）。12.7%的活跃率凸显了策略的筛选特性，严格遵循其保守的Z分数入场标准。显著下降的胜率——从训练阶段的30%降至测试阶段的6.7%——表明评估期间均值回归模式明显减弱。这种表现差异凸显了保守参数设置与严谨风险管控对于策略持久性的重要性。

3.5 PLAN

3.5 计划

As a counter-trend approach, Strategy S3 (Mean Reversion) remains vulnerable to substantial losses during strong directional trends, as evidenced in its Data 1 performance. To address this specific weakness, the Semester 2 development plan will implement an ADX-based market regime filter. Unlike the S1 implementation, we will test the rule: "only execute mean-reversion trades when ADX readings fall below a specified threshold (e.g., 20)." This criterion specifically targets low-trending, range-bound market conditions that match the strategy's ideal operating parameters. Our development roadmap begins with initial testing on Data 1 to establish baseline performance, then shifts core development to Data 2. We will divide this dataset into separate training and out-of-sample validation sets, using the training portion to recalibrate the ADX threshold and the OOS segment to verify the filter's effectiveness in avoiding unfavorable trending periods within the new market context.

作为一种逆势策略，S3策略（均值回归）在强劲趋势期间仍易遭受重大损失，这一点在其Data 1的表现中得到了证实。为解决这一特定弱点，第二学期的开发计划将实施基于ADX的市场机制过滤器。与S1的实施不同，我们将测试以下规则：“仅在 ADX 读数低于特定阈值（例如20）时执行均值回归交易。”该标准特别针对低趋势、区间波动的市场条件，这与策略的理想运行参数相匹配。我们的开发路线图首先在Data 1上进行初步测试以建立基线表现，然后将核心开发转移到Data 2。我们将该数据集分为独立的训练集和样本外验证集，使用训练部分重新校准 ADX 阈值，并使用OOS部分验证过滤器在新市场环境中避免不利趋势期的有效性。

PART 2: COMBINED CVAR STRATEGY DESIGN

第2部分：联合 CVAR 策略设计

STRATEGY COMBINATION METHODOLOGY

战略组合方法论

The combined strategy implements an advanced Conditional Value at Risk (CVaR) optimization framework integrating all three individual strategies with dynamic weight allocation. Starting with £1,000,000 in total capital, the system employs CVaR optimization to determine optimal portfolio weights every 20 trading days based on a 60-day historical window. The 10% minimum and 70% maximum allocation limits for each strategy effectively balance diversification with optimization flexibility. Each strategy operates exclusively within its designated market series, maintaining independent entry and exit logic while the portfolio-level optimization system manages overall capital allocation across all strategies.

这套组合策略采用先进的条件价值风险（CVaR）优化框架，通过动态权重分配整合三种独立策略。系统初始总资本为100万英镑，每20个交易日基于60天历史数据窗口，运用CVaR优化算法确定最优投资组合权重。每项策略均设有10%的最低配置和70%的最高配置限制，有效平衡了分散投资与优化灵活性。各策略独立运作于指定市场系列，保持独立的入场和离场逻辑，而投资组合层面的优化系统则统筹管理所有策略的资本配置。

The CVaR optimization process minimizes the expected loss in the worst 5% of outcomes by analyzing daily returns for each strategy, calculating portfolio CVaR at the 95% confidence level, and adjusting weights within constraints to minimize tail risk. This approach provides superior downside protection compared to traditional mean-variance optimization while maintaining competitive returns.

CVaR优化流程通过分析各策略每日收益、计算95%置信水平下的组合CVaR，并在约束条件下调整权重以最小化尾部风险，从而最小化最差5%结果的预期损失。与传统均值-方差优化相比，该方法在保持竞争力回报的同时，提供了更优越的下行保护。

[10]

[10]

4.1 CVAR MATHEMATICAL FRAMEWORK

4.1 CVAR 数学框架

Conditional Value at Risk (CVaR), also known as Expected Shortfall, quantifies the expected loss given that the loss exceeds the Value at Risk (VaR) threshold. For a portfolio with return distribution R and confidence level α, CVaR is defined as:

条件风险价值（CVaR），也称为预期短缺，量化了损失超过风险价值（VaR）阈值时的预期损失。对于具有收益分布R和置信水平 α 的投资组合，CVaR定义为：

$$\text{VaR}_\alpha = \inf\{r \in \mathbb{R}: P(R \le r) \ge \alpha\}$$

$$\text{VaR}_\alpha = \inf\{\text{r} \in \mathbb{R}：\ \text{P（R} \le \text{r）} \ge \alpha\}$$

$$\text{CVaR}_\alpha = E[R \mid R \le \text{VaR}_\alpha] = \frac{1}{\alpha}\int_0^\alpha \text{VaR}_\beta \, d\beta$$

$$\text{CVaR}_\alpha = E[R \mid R \le \text{VaR}_\alpha] = \frac{1}{\alpha}\int_0^\alpha \text{VaR}_\beta \, d\beta$$

For a portfolio of n strategies with weights $w = (w_1, \ldots, w_n)$ and returns r_i, the portfolio return is:

对于一个包含n个策略的投资组合，其权重为w=（$w_1, \ldots, w_n$），收益为r_i，投资组合收益为：

$$R_p = \sum_{i=1}^{n} w_i \, r_i$$

$$R_p = \sum_{i=1}^{n} w_i \, r_i$$

The optimization problem minimizes CVaR subject to portfolio constraints:

该优化问题在投资组合约束条件下最小化条件价值风险（CVaR）：

$$\min_w \text{CVaR}_\alpha(R_p)$$

$$\text{最小权重CVaR}_\alpha（R_p）$$

Subject to:

但须遵守：

$$\sum_{i=1}^{n} w_i = 1$$

$$\sum_{i=1}^{n} w_i = 1$$

$$w_{\min} \le w_i \le w_{\max}, \quad \forall i \in \{1, \dots, n\}$$

In this implementation, $\alpha = 0.05$ (95% confidence), $w_{\min} = 0.10$, and $w_{\max} = 0.70$.

$$w_{\min} \le w_i \le w_{\max}, \quad \forall i \in \{1, \dots, n\}$$

. , n}。在此实现中， = 0.05（95%置信度），$w_{\min}$=0.10，$w_{\max}$= 0.70。

4.2 SLSQP OPTIMIZATION ALGORITHM [6]

4.2 SLSQP 优化算法[6]

The Sequential Least Squares Programming (SLSQP) algorithm is a gradient-based iterative method employed to solve the constrained CVaR optimization problem. At each iteration, the algorithm constructs a quadratic programming subproblem by approximating the objective function with a quadratic model and linearizing the constraints. This subproblem is then solved to determine a search direction, which is combined with a line search to update the portfolio weights. This iterative process continues until the convergence criteria are satisfied.

顺序最小二乘规划（SLSQP）算法是一种基于梯度的迭代方法，用于解决约束条件下的CVaR优化问题。在每次迭代中，该算法通过用二次模型近似目标函数并线性化约束条件，构建一个二次规划子问题。然后求解该子问题以确定搜索方向，再结合线性搜索更新投资组合权重。这一迭代过程持续进行，直到满足收敛标准。

SLSQP offers several advantages, including its ability to efficiently manage both equality and inequality constraints, its robust convergence properties, and its computational efficiency. The implementation utilizes **scipy.optimize.minimize** with method='SLSQP', which provides automatic gradient estimation and constraint handling.

SLSQP 有以下几个优点，包括能够高效管理等式和不等式约束、稳健的收敛特性和计算效率。该实现使用**scipy.optimize.minimize** with method=‘ SLSQP ’，它提供了自动梯度估计和约束处理。

4.3 CVAR OPTIMIZATION PSEUDOCODE

4.3 CVAR 优化伪代码

```
Algorithm: CVaR Portfolio Optimization
CVaR投资组合优化算法

Input:
输入：
  - returns_matrix: Historical returns for n strategies (T × n matrix)
  - returns_matrix：n种策略的历史收益率（T×n矩阵）
  - lookback_window: Number of historical days for optimization (60)
  - 回溯窗口：用于优化的历史天数（60）
  - rebalance_frequency: Days between rebalancing (20)
  - rebalance_frequency：再平衡间隔天数（20）
  - confidence_level: CVaR confidence level α (0.05)
  - confidence_level：CVaR置信水平 （0.05）
  - min_weight, max_weight: Portfolio weight constraints (0.10, 0.70)
  - min_weight，max_weight：投资组合权重约束（0.10,0.70）

Output:
输出：
  - optimal_weights: Portfolio allocation weights
  - 最优权重：投资组合配置权重

Initialize:
初始化：
  weights = [1/n, 1/n, ..., 1/n]  // Equal initial allocation private static int MAX = 5;
  权重 = [1/n，1/n ，...，1/n] // 等等的初始分配私有静态int MAX = 5；

For each trading day t:
对于每个交易日t：
  If t mod rebalance_frequency == 0:
  若t mod rebalance_frequency == 0：

    // Extract rolling window of returns
    // 提取滚动窗口的返回值
    window_returns = returns_matrix[t-lookback_window:t, :]
    window_returns = returns_matrix[t-lookback_window:t, :]

    // Define CVaR objective function
    // 定义CVaR目标函数
    Function calculate_cvar(w):
    函数计算标准差(w)：
      portfolio_returns = window_returns @ w var_threshold = percentile(portfolio_returns, α × 100)
      portfolio_returns = window_returns @ w var_threshold = percentile（portfolio_returns， Γ × 100）
      cvar = mean(portfolio_returns[portfolio_returns ≤ var_threshold])
      cvar = 均值（投资组合收益率[投资组合收益率 ≤ 变量阈值]）

      Return cvar
      返回变量

// Define optimization constraints
  constraints = [
    sum(w) == 1,          // Weights sum to 1
    和(w)等于1，          // 权重之和为1
    w[i] >= min_weight,   // Minimum allocation
    w[i] >= min_weight，// 最小分配值
    w[i] <= max_weight      // Maximum allocation ]
    当w[i] <= 最大权重时

// Solve optimization problem using SLSQP
// 使用 SLSQP 解决优化问题
optimal_weights = SLSQP_minimize( objective = calculate_cvar, initial_guess = weights, constraints = constraints,
最优权重 = SLSQP最小化（目标函数 = 计算方差，初始猜测 = 权重，约束条件 = 约束条件）
  bounds = [(min_weight, max_weight) for i in 1:n]
  边界条件 = [（最小权重，最大权重）对于 i 在 1：n]
)
)

weights = optimal_weights
权重 = 最优权重

// Apply current weights to determine position sizes
// 应用当前权重以确定位置大小
For each strategy i:
对于每种策略i：
  allocated_capital[i] = total_capital × weights[i] strategy[i].set_capital(allocated_capital[i])
  allocated_capital[i] = total_capital × weights[i] strategy[i].set_capital(allocated_capital[i])

Return weights
回归权重
```

4.4 CVAR VS MEAN-VARIANCE OPTIMIZATION COMPARISON [9,10,11]

4.4 CVAR 与均值-方差优化比较[9,10,11]

Traditional Mean-Variance (MV) optimization [9] aims to minimize total portfolio variance $\sigma_p^2 = w^T \Sigma w$, thereby penalizing both upside (favorable) and downside (unfavorable) volatility equally. This approach assumes returns are normally distributed [3], a premise that often fails to capture the "fat-tailed" distributions (i.e., extreme events) common in financial markets.

传统均值-方差（MV）优化[9]旨在最小化投资组合总方差$\sigma_p^2 = w^T \Sigma w$, 该方法对市场波动的上行（有利）和下行（不利）风险均施加同等惩罚。其理论基础是假设收益呈正态分布[3]，但这一前提往往无法反映金融市场中常见的‘肥尾’分布（即极端事件）。

In contrast, CVaR optimization [10] focuses exclusively on downside tail risk, seeking to minimize the expected loss in the worst-case scenarios (e.g., the worst 5% of outcomes). This asymmetric risk measure aligns better with investor preferences to avoid significant losses. As a coherent risk measure [10], CVaR is theoretically superior and effectively handles the non-normal and fat-tailed return distributions that MV models struggle with.

相比之下，条件风险价值（CVaR）优化[10]专注于下行尾部风险，旨在最小化最坏情况（如最差5%结果）下的预期损失。这种非对称风险度量更符合投资者规避重大损失的偏好。作为一致性风险度量[10]，CVaR在理论上更具优势，能有效处理均值风险（MV）模型难以应对的非正态和厚尾收益分布。

For this portfolio, the CVaR approach provided robust protection against extreme events and achieved superior risk-adjusted returns (Annualized Sharpe Ratio [11] of 1.061).

对于该投资组合，CVaR方法提供了对极端事件的稳健保护，并实现了更优的风险调整回报（年化夏普比率[11]为1.061）。

4.5 PORTFOLIO WEIGHT EVOLUTION
4.5 投资组合权重演变

CVAR DYNAMIC REBALANCING PROCESS
CVAR 动态再平衡过程

The portfolio management approach utilizes periodic rebalancing at 20-trading-day intervals through CVaR optimization. The process commences with an initial equal-weight allocation (33. 3% per strategy) across all three approaches. Subsequently, leveraging a 60-day rolling returns window, the optimizer determines revised allocations to minimize tail risk, specifically targeting Conditional Value at Risk (CVaR) at the 5% significance level. This tail risk optimization process operates within defined constraints: each strategy's allocation remains between a 10% minimum and 70% maximum to preserve diversification and prevent over concentration. The Sequential Least Squares Programming (SLSQP) algorithm provides the mathematical foundation for achieving this optimization objective.

投资组合管理方法通过条件价值风险（CVaR）优化，在20个交易日的间隔内进行周期性再平衡。该过程首先以初始等权重分配（每种策略33.3%）覆盖所有三种方法。随后，利用60天滚动回报窗口，优化器确定修订后的分配，以最小化尾部风险，具体目标是在5%显著性水平下实现条件价值风险（CVaR）最小化。这一尾部风险优化过程在定义的约束内进行：每种策略的分配保持在10%最低和70%最高之间，以保持分散化并防止过度集中。顺序最小二乘规划（SLSQP）算法为实现这一优化目标提供了数学基础。

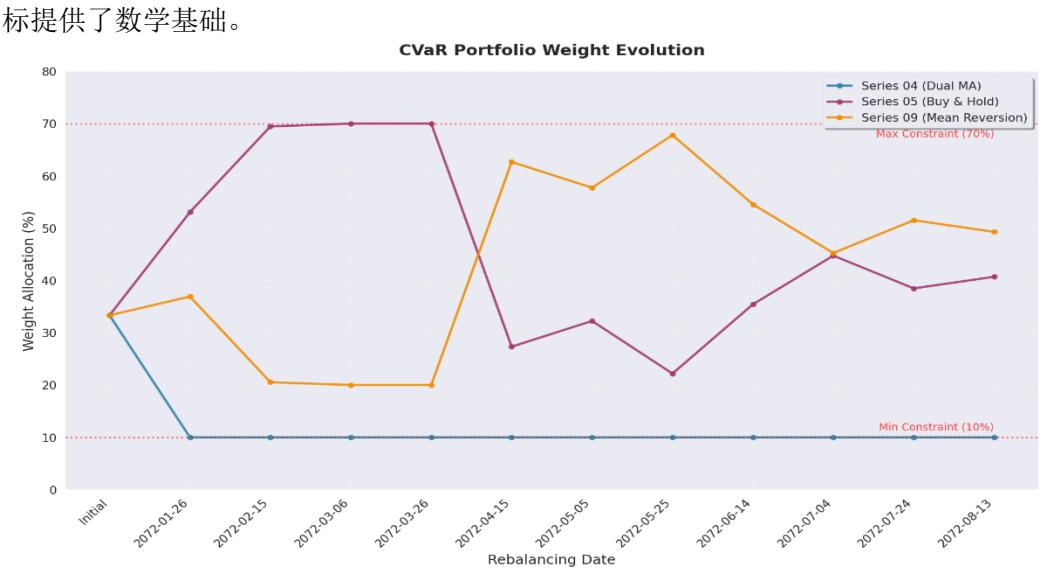


Figure 4.1: CVaR portfolio weight evolution during out-of-sample test period (Days 701-1000)

图4.1：CVaR投资组合权重在样本外测试期间（第701-1000天）的演变

Weight Allocation Statistics (Test Period Only)

权重分配统计（仅限测试期）

Strategy 战略	Initial 初始	Final 最后	Mean 平均	Std Dev 标准差	Min 最小值	Max 最高的
Series 04 (Dual MA) 系列04（双MA）	33.3% 33.3%	10.0% 10.0%	11.9% 11.9%	6.4% 6.4%	10.0% 10.0%	33.3% 33.3%
Series 05 (Buy & Hold) 05系列（买入并持有）	33.3% 33.3%	40.7% 40.7%	44.8% 44.8%	16.3% 16.3%	22.2% 22.2%	70.0% 70.0%
Series 09 (Mean-reversion) 系列09（均值回归）	33.3% 33.3%	49.3% 49.3%	43.3% 43.3%	16.2% 16.2%	20.0% 20.0%	67.8% 67.8%

KEY OBSERVATIONS (TEST PERIOD):
关键观察结果（测试期）：

Throughout the testing period, the allocation to Series 04 (Dual MA) consistently maintained the minimum 10% constraint following the initial rebalance, reflecting its comparatively higher tail risk contribution relative to the other strategies. This minimum allocation persisted across all 11 rebalancing events. In contrast, Series 05 (Buy & Hold) achieved the highest average allocation at 44.8%, reaching the maximum 70% constraint during February and March 2072 amid robust trending performance—demonstrating its superior risk-adjusted returns and lower CVaR impact. Series 09 (Mean-reversion) exhibited the most volatile allocation pattern with a standard deviation of 16.2%, undergoing dramatic shifts between 20% and 67.8%. This strategy received its peak allocations (62. 7%-67.8%) during April to May 2072 when market conditions particularly favored mean-reversion approaches. These allocation dynamics reveal two distinct regime shifts: an initial phase (January-March 2072) dominated by Series 05 with 53%-70% allocations during trending markets, followed by a mid-to-late period (April-August 2072) where Series 09 gained prominence with 45%-68% allocations as markets transitioned to range-bound conditions.

在整个测试期间，04系列（双重MA）在初始再平衡后始终维持最低10%的配置限制，这反映了其相对于其他策略较高的尾部风险贡献。这一最低配置在所有11次再平衡事件中都保持不变。相比之下，05系列（买入并持有）实现了最高的平均配置比例，达到44.8%，并在2072年2月和3月期间达到了70%的最大配置限制，表现出强劲的趋势性表现——这证明了其风险调整后的回报率更高且CVaR影响更低。09系列（均值回归）则表现出最波动的配置模式，标准差为16.2%，在20%和67.8%之间经历了剧烈波动。该策略在2072年4月至5月期间获得了最高配置（62.7%-67.8%），当时市场条件特别有利于均值回归策略。这些配置动态呈现出两种截然不同的市场格局转变：在趋势性市场中，05系列占据主导地位，配置比例达53%-70%（2072年1-3月）；随后进入中期至后期阶段（2072年4-8月），随着市场转向区间波动，09系列逐渐占据主导地位，配置比例升至45%-68%。

RISK MANAGEMENT FRAMEWORK
风险管理框架

5.1 INDIVIDUAL TRADE RISK MANAGEMENT
5.1 个体贸易风险管理

For Strategy S1 (Dual MA), a 10% stop-loss combined with a 20% take-profit creates an asymmetric 2:1 risk-reward structure appropriate for trend-following methodologies, with these thresholds calibrated according to Series 04's historical volatility patterns. Strategy S2 (Buy & Hold) intentionally omits intra-trade stop-loss mechanisms—a deliberate design choice to prevent premature exit from long-term uptrends. Instead, its risk management operates through rigorous initial asset selection and portfolio-level capital allocation rather than tactical position exits. Meanwhile, Strategy S3 (Mean-reversion) implements a tight 5% stop-loss alongside a 4% take-profit, reflecting its contrarian nature where unsuccessful signals require prompt termination. This framework is further

对于策略S1（双移动平均线策略），10%的止损与20%的止盈相结合，形成了适合趋势跟踪方法的2：1风险收益不对称结构，这些阈值根据04系列的历史波动模式进行校准。策略S2（买入并持有）刻意省略了交易中的止损机制——这是为防止在长期上涨趋势中过早退出而做出的策略性设计。其风险管理通过严格的资产筛选和组合层面的资金配置来实现，而非依赖战术性仓位退出。与此同时，策略S3（均值回归策略）采用5%的严格止损与4%的止盈组合，体现了其逆向操作特性：当信号失效时需立即终止。该框架进一步

reinforced by a 20-day maximum holding period constraint, preventing extended capital commitment to positions that fail to revert within expected timeframes.
通过设定20天的最长持仓期限限制来强化约束，防止对未能在预期时间内平仓的头寸进行长期资本承诺。

5.2 PORTFOLIO-LEVEL RISK MANAGEMENT
5.2 组合风险控制

DYNAMIC WEIGHT REBALANCING
动态配重

Every 20 trading days, the CVaR optimizer recalculates optimal weights using the rolling 60-day return window. This automatic rebalancing mechanism adapts to changing market conditions and relative strategy performance without manual intervention. Weight constraints (10%-70% per strategy) ensure meaningful diversification while allowing sufficient flexibility for optimization.
CVaR优化器每20个交易日会基于滚动60天收益窗口重新计算最优权重。这种自动再平衡机制无需人工干预，即可适应市场环境变化和策略表现波动。通过设置每策略10%-70%的权重限制，既保证了有效分散投资，又为优化操作留有充足空间。

UNDERPERFORMING STRATEGY ADAPTATION
UNDERPERFORMING 战略调整

The CVaR optimization framework automatically reduces allocation toward the 10% minimum for strategies showing persistent underperformance. This process uses a rolling 60-day window to detect performance deterioration, gradually decreasing weights for strategies with negative Sharpe ratios or elevated CVaR readings. The system reallocates capital through 20-day rebalancing cycles, avoiding sudden portfolio changes while distinguishing between temporary setbacks and fundamental strategy breakdown—all without manual intervention. A 10% minimum allocation is preserved even for chronically underperforming strategies, maintaining diversification benefits and allowing potential recovery during favorable regime shifts. This automated process operates without discretionary intervention, eliminating behavioral biases while preserving systematic risk discipline. For instance, should Strategy S3 (Statistical Mean-reversion) generate persistent losses during strong trending markets, the optimizer would identify this performance deterioration through the rolling window. Across subsequent rebalancing cycles, S3's allocation would decrease toward the 10% floor while S1 (Dual Moving Average Crossover) and S2 (Buy and Hold) strategies receive increased weights better suited to the prevailing market regime.
CVaR优化框架能自动将持续表现不佳的策略配置比例自动调低至10%的最低标准。该机制采用滚动60天窗口检测业绩恶化，对夏普比率为负或CVaR值偏高的策略逐步降低权重。系统通过20天的周期性再平衡调整资本配置，既能避免投资组合的剧烈波动，又能区分短期震荡与策略根本性失效——整个过程无需人工干预。即使长期表现欠佳的策略仍保持10%的最低配置比例，既维护了资产分散化优势，又为市场环境转好时的潜在复苏留有余地。这套自动化流程无需人工干预，既消除了行为偏差，又保持了系统性风险管理的严谨性。例如，当统计均值回归策略S3在强势市场中持续亏损时，优化器会通过滚动窗口及时识别这种业绩恶化。在后续的再平衡周期中，S3的配置比例将逐步向10%的下限靠拢，而S1（双移动平均线交叉策略）和S2（买入并持有策略）则会获得更优的权重调整，以更好地适应当前的市场环境。

5.3 CROSS-STRATEGY AND CROSS-SERIES RISK MANAGEMENT
5.3 跨战略、跨系列风险管理

CORRELATION MANAGEMENT
相关管理

CVaR optimization automatically addresses correlation risk through its return-based optimization framework. Strategies operating on different underlying series naturally benefit from inherent diversification due to their lower correlation structure compared to multiple strategies trading the same instrument. The CVaR minimization process further enhances this structural benefit by systematically favoring strategy combinations that demonstrate reduced joint tail risk exposure.
CVaR优化通过其基于收益的优化框架自动化解相关性风险。与交易同一金融工具的多策略相比，基于不同基础序列运作的策略因其较低的相关性结构，天然具备分散化优势。CVaR最小化过程通过系统性地偏好那些能降低联合尾部风险敞口的策略组合，进一步强化了这种结构性优势。

MARKET REGIME ADAPTATION
市场体制适应

The portfolio framework dynamically adjusts to evolving market conditions through its strategic allocation mechanism. During pronounced trending periods, allocations naturally favor S1 (Dual MA) and S2 (Buy & Hold) strategies, while ranging markets trigger increased weighting toward S3 (Mean-reversion). In transitional or mixed regimes, capital becomes more evenly distributed across all three approaches. This adaptive process operates automatically through the rolling window methodology, continuously aligning portfolio composition with prevailing market dynamics without requiring explicit regime classification or manual intervention.
该投资组合框架通过其战略配置机制，能够动态适应不断变化的市场环境。在趋势明显时期，系统会自然倾向采用S1（双移动平均线）和S2（买入并持有）策略；当市场处于震荡区间时，则会加大S3（均值回归）策略的权重。在市场过渡期或混合策略阶段，资金会在三种策略间实现更均衡的分配。这种自适应机制通过滚动窗口方法自动运行，无需人工分类或干预，即可持续将投资组合结构与当前市场动态保持同步。

5.4 BANKRUPTCY RISK MITIGATION
5.4 破产风险的缓解

FRAMEWORK-LEVEL PROTECTIONS
框架级 保护

The framework incorporates multiple automated safeguards to ensure robust bankruptcy protection. The system employs pre-trade expenditure validation, where all pending market orders undergo cash balance verification before execution; any transaction that would result in a negative cash position is automatically rejected, effectively preventing unauthorized borrowing or margin exposure.Additionally, strict capital allocation compliance ensures each strategy operates within its predetermined CVaR-optimized weighting,eliminating the possibility of individual strategy over-leverage that could compromise overall portfolio solvency. Finally, continuous real-time net worth monitoring (cash + position valuations) provides a final protective layer: should the total portfolio value decline below zero, immediate automatic liquidation of all positions at prevailing market prices is triggered, followed by complete trading suspension.
该框架通过多重自动化防护机制构建起完善的破产保护体系。系统采用交易前资金验证机制，所有待执行的市场订单在成交前都会进行现金余额核验；任何可能导致资金缺口的交易都会被自动拦截，有效防范未经授权的借贷或保证金风险。此外，严格的资本配置合规机制确保每个策略都严格遵循预先设定的CVaR优化权重，杜绝因单一策略过度杠杆化而危及整体投资组合偿付能力的风险。最后，持续实时的净资产监控（现金+头寸估值）形成最后一道防线：当投资组合总价值跌破零点时，系统将立即按现行市场价格自动清算所有头寸，并全面暂停交易。

STRATEGY-LEVEL PROTECTIONS
战略层面的保护措施

Strategy-level protections are enforced through several key mechanisms. Individual trade stop-loss disciplines (10% for S1, 5% for S3) limit the maximum loss per position; these stop-losses are implemented at the strategy level to enforce risk discipline regardless of portfolio-level optimization. Furthermore, position sizing constraints mandate individual strategies size positions as percentages of their allocated capital (25% for S1, 100% for S2, 15% for S3) rather than total portfolio value. This ensures no single trade can
策略层面的风控机制通过三大核心手段实现。首先，单笔交易设置止损纪律（S1策略10%，S3策略5%），将每笔交易的最大亏损限制在特定比例内。这些止损规则在策略层面强制执行，确保风险管控不因投资组合优化而松懈。其次，仓位管理约束要求各策略必须按资本配置比例（S1策略25%，S2策略100%，S3策略15%）来确定持仓规模，而非依据投资组合总值。这种机制能有效防止任何单一交易操作

consume excessive capital even within the strategy's allocation. Finally, the 10%-70% weight constraints function as diversification requirements, preventing over-concentration in any single strategy and ensuring that bankruptcy in one strategy component cannot propagate to the entire portfolio.
即便在策略配置范围内，仍存在过度资本消耗。最后，10%-70%的权重限制作为分散化要求，既防止单一策略过度集中，又确保某策略组件的破产不会波及整个投资组合。

CVAR-SPECIFIC BANKRUPTCY PROTECTION
CVAR-特定破产保护

CVaR optimization inherently delivers bankruptcy protection through systematic tail risk minimization. By concentrating on the worst 5% of potential outcomes, the optimization algorithm systematically reallocates capital away from strategies demonstrating significant losses or heightened downside volatility. This focus on extreme loss scenarios directly addresses the conditions that typically precede insolvency events, establishing a proactive rather than reactive protective mechanism.
CVaR优化通过系统性尾部风险最小化，本质上实现了破产保护。该算法聚焦于最糟糕的5%潜在结果，将资本系统性地从高亏损或下行波动剧烈的策略中转移。这种对极端亏损场景的聚焦，直接应对了通常引发破产事件的条件，从而建立了一种主动而非被动的保护机制。
Bankruptcy Probability Analysis: Given the portfolio's maximum drawdown of 3.86% and Sharpe ratio of 1.061 across the entire backtesting period, the probability of complete capital loss (100% impairment) remains statistically insignificant under normal market conditions. The comprehensive protection system—featuring diversified strategy exposure, conservative position sizing, CVaR-based tail risk management, and automated rebalancing—collectively forms a robust defense against catastrophic loss events.
破产概率分析：在回测期间，该投资组合的最大回撤率为3.86%，夏普比率达1.061。在正常市场环境下，完全资本损失（100%减值）的概率在统计学上仍不显著。这套综合防护体系——包含分散化策略布局、保守仓位管理、基于CVaR的尾部风险管理以及自动再平衡机制——共同构筑起抵御灾难性损失事件的坚实防线。

POSITION SIZING METHODOLOGY
位置尺寸确定方法

The CVaR optimization framework establishes strategic allocations by drawing upon historical return distributions, correlation patterns, tail risk profiles, and risk-adjusted performance metrics. Within these predetermined allocations, the individual strategies operate under specific position sizing rules: the dual MA strategy commits 25% of allocated capital per trade, buy-and-hold invests 100% in a single position, and mean-reversion allocates 15% per trade. Portfolio rebalancing takes place every 20 trading days, activated once enough fresh data becomes available for meaningful optimization. The execution process seamlessly transitions positions toward new target weights, maintaining portfolio efficiency while minimizing market impact.
CVaR优化框架通过综合运用历史收益分布、相关性模式、尾部风险特征及风险调整绩效指标，构建出战略性资产配置方案。在这些预设配置框架下，各投资策略遵循特定的仓位管理规则：双移动平均策略每笔交易投入25%的配置资金，买入并持有策略将全部资金集中于单一仓位，均值回归策略每笔交易配置15%资金。每20个交易日进行一次投资组合再平衡，当积累足够新鲜数据用于有效优化时即触发该机制。执行过程中，系统会无缝切换仓位至新目标权重，在保持投资组合效率的同时，最大限度降低市场波动影响。

COMBINED STRATEGY PERFORMANCE RESULTS
综合战略绩效结果

Performance Metric 性能指标	Value 值
Initial Capital 初始资本	£1,000,000.00 £1,000,000.00
Final Portfolio Value 最终投资组合价值	£1,129,321.25 £1,129,321.25
Total Return 总回报	+12.93% +12.93%
Annualized Return 年化收益率	+3.12% +3.12%
Annualized Sharpe Ratio 年化夏普比率	1.061 1.061
Maximum Drawdown 最大回撤	3.86% 3.86%
PD Ratio PD比值	3.35 3.35
Win Rate 赢率	50.3% 50.3%
Activity Level 活动水平	93.9% 93.9%
Trading Days 交易日	999 999
Bankruptcy Events 破产事件	0 0

Table 7.1: CVaR Combined Strategy Performance(Full Period Days 1-1000)
表7.1：CVaR组合策略绩效（完整周期第1-1000天）

The CVaR-optimized portfolio achieved exceptional risk-adjusted results, producing a 1.061 annualized Sharpe ratio that outperformed a simple equal-weight benchmark. Its 3. 86% maximum drawdown confirms successful tail risk control through systematic CVaR implementation. The portfolio's 3.35 return-to-drawdown ratio demonstrates strong performance relative to potential losses, validating the dynamic weighting approach. Additionally, the 93.9% activity level indicates consistent management involvement across all three asset series during the testing period.
经过CVaR优化的投资组合展现出卓越的风险调整收益，其年化夏普比率高达1.061，明显优于简单的等权重基准。3.86%的最大回撤率证明，通过系统化的CVaR管理策略，尾部风险得到有效控制。3.35的回撤回报率表明，该组合在潜在损失面前表现稳健，充分验证了动态加权策略的有效性。此外，93.9%的活跃度指标显示，在测试期间所有三个资产系列均保持了持续的管理参与。

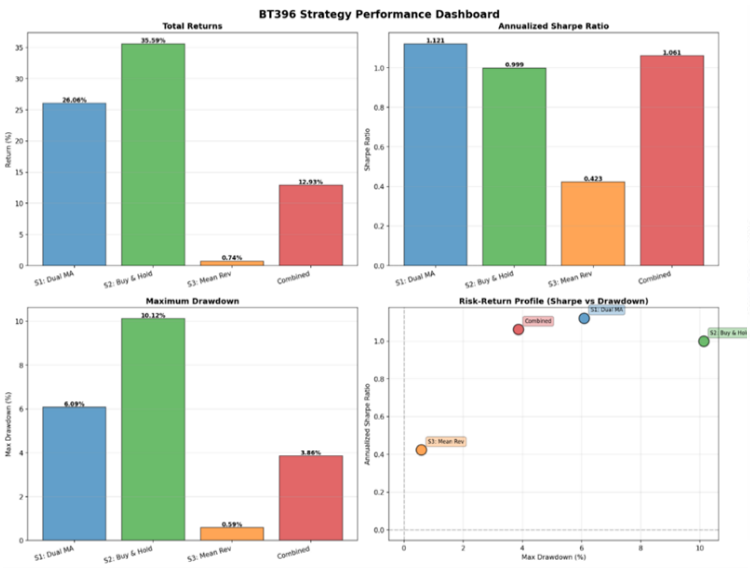


Figure 7.1: Cumulative return comparison for all four strategies over full backtest period

图7.1：全部四种策略在完整回溯测试期间的累积回报比较

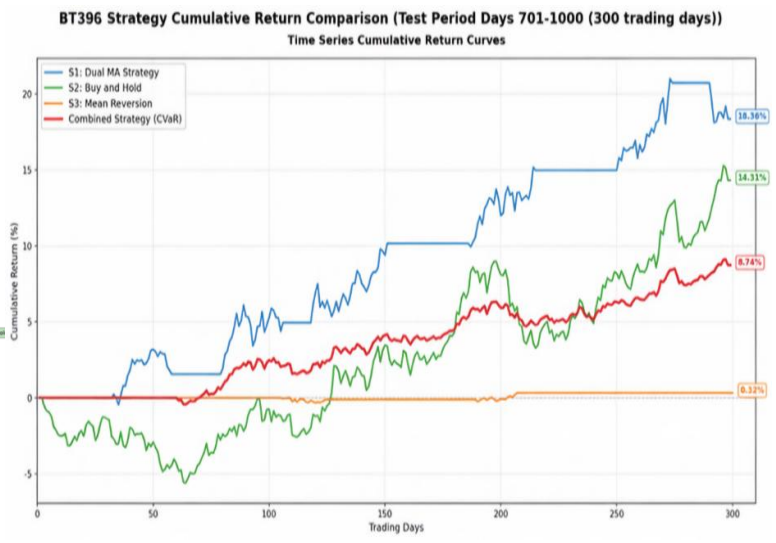


Figure 7.2: Comprehensive performance dashboard comparing returns, Sharpe ratios, drawdowns, and risk-return profiles.

图7.2：综合绩效仪表板，比较了回报率、夏普比率、回撤率和风险回报曲线。

The CVaR-optimized portfolio effectively balanced exposure across three distinct strategies, delivering superior risk-adjusted returns through dynamic reallocation guided by tail risk minimization. With an annualized Sharpe ratio of 1.061 and total returns of 12.93%, the results demonstrate the efficacy of CVaR-based portfolio management in multi-strategy integration. Moreover, the contained maximum drawdown of 3.86% validates the robustness of the tail risk management approach throughout the complete backtesting period.

经过CVaR优化的投资组合在三种不同策略间实现了有效风险分散，通过动态再配置策略（以尾部风险最小化为指导）实现了卓越的风险调整回报。该组合年化夏普比率达1.061，总回报率高达12.93%，充分证明了基于CVaR的多策略组合管理方案的卓越成效。更值得关注的是，其最大回撤率仅3.86%，这一数据充分验证了尾部风险管理策略在整个回测周期内的稳健性。

PLAN
计划

The Semester 2 development for the combined strategy will be twofold. First, we will improve the quality of its constituents by integrating the three enhanced individual strategies (S1+ADX, S2+200DMA, S3+ADX). This will make the inputs for our CVaR optimizer (i.e., the individual strategy returns) more robust, leading to a more adaptive combined portfolio. Second, we will address a key execution risk: our current reliance on Market Orders, which exposes the portfolio to significant slippage risk from "High Jumps" or overnight price gaps. We will research and test the use of Limit Orders for trade entry to control our execution price rather than being subject to potentially adverse opening prices. For data usage, Data 1 will serve as the environment for preliminary research and tool development (e.g., initial testing of the Limit Order execution logic). However, the final parameter calibration and combined portfolio optimization must be performed on Data 2. We will partition Data 2 into a new training set (e.g., Days 101-700) and a final OOS test set (e.g., Days 701-1000). The CVaR optimizer will be run and recalibrated on the Data 2 training set, and the OOS test set will serve as the definitive test to validate if the upgraded system achieves superior risk-adjusted returns and if the Limit Order logic successfully reduced slippage costs in the new environment.

第二学期的联合策略开发将分为两个方面。首先，我们将通过整合三个增强的独立策略（S1+ ADX 、S2+200DMA、S3+ADX）来提高其组成部分的质量。这将使我们的CVaR优化器输入（即独立策略收益）更加稳健，从而形成更具适应性的组合投资组合。其次，我们将解决一个关键的执行风险：目前我们依赖市场订单，这使得投资组合面临“高跳”或隔夜价格缺口带来的显著滑点风险。我们将研究并测试使用限价单进行交易以控制执行价格，而不是受潜在不利开盘价格的影响。对于数据使用，数据1将作为初步研究和工具开发的环境（例如，对限价单执行逻辑的初步测试）。然而，最终的参数校准和组合投资组合优化必须在数据2上进行。我们将数据2划分为新的训练集（例如，第101-700天）和最终的离线测试集（例如，第701-1000天）。CVaR优化器将在Data 2训练集上运行并重新校准，而OOS测试集将作为最终验证，以确认升级后的系统是否实现了更优的风险调整回报，以及限价单逻辑是否成功降低了新环境下的滑点成本。

PART 3: BREAKDOWN OF THE TEAMWORK
第三部分：团队合作的分解

ZHAO YIXUAN
ZHAO YIXUAN

1. Background Research & Asset Selection

1. 背景研究与资产选择

During the background research and asset selection phase, I assisted team member (Zhang Ao) with the analysis and plotting of the price evolution charts for all 10-asset series (i.e., Fig 0.1 in the report). Additionally, I wrote scripts to perform the project's early Exploratory Data Analysis (EDA), contributing key statistical visuals including the **correlation matrix, volatility analysis, and Hurst exponent comparisons**.

在背景调研和资产筛选阶段，我协助团队成员张奥完成了全部10个资产系列的价格走势图分析与绘制（即报告中的图0.1）。此外，我还编写了脚本进行项目的早期探索性数据分析（EDA），贡献了**关键统计可视化成果，包括相关性矩阵、波动性分析以及赫斯特指数对比**。

2. Coding & Strategy Implementation

2. 编码与策略的实现

I was responsible for writing all Python scripts for the project, transforming all team members' proposed strategic concepts into functional Backtrader code.

我负责为该项目编写所有Python脚本，将团队成员提出的所有战略概念转化为Backtrader功能代码。

- **S1: Dual MA Strategy:** Following discussions with team member (Zhang Ao) who proposed the trend-following strategy, I handled the technical implementation by writing s1_dual_ma_crossover.py. The fundamental logic of this strategy is: within the next method, it continually monitors the state of a 3-day fast MA and a 24-day slow MA. When the fast MA crosses above the slow MA (self.crossover > 0), a buy signal is triggered. While in a position, the code contemporaneously checks if
- **S1：双MA策略：**在与团队成员张奥（Zhang Ao）讨论趋势跟踪策略后，我通过编写s1_dual_ma_crossover.py代码实现了该策略的技术实现。该策略的核心逻辑是：在下方方法中，系统会持续监测3日快速移动平均线（MA）和24日慢速移动平均线（MA）的状态。当快速MA突破慢速MA上方时（自动交叉>0），系统将触发买入信号。在持仓期间，代码会同步检查是否

the price hits the 10% stop-loss or 20% take-profit levels, automatically liquidating the position to control risk. The core parameters are defined as:

当价格触及10%止损位或20%止盈位时，系统将自动平仓以控制风险。核心参数定义如下：

- **S2: Buy and Hold Strategy:** Responding to the Buy and Hold strategy concept proposed by team member (Haw), I was responsible for the full code implementation in s2_buy_and_hold.py. The logic ensures a one-time purchase: a self.bought = False flag is set in the next method, permitting a single buy order on the first available day. The code calculates and purchases the maximum number of shares possible with the available cash and then holds this position until the end of the backtest.
- **S2：买入并持有策略：**针对团队成员（Haw）提出的买入并持有策略概念，我负责在s2_buy_and_hold.py中完成全部代码实现。逻辑设计确保单次买入：在后续方法中设置一个self bought = False False旗标志，允许在首个可用交易日进行单次买入指令。代码会根据可用资金计算并买入最大可行股数，随后持续持有该仓位直至回测结束。
- **S3: Mean Reversion Strategy:** Based on the mean-reversion concept proposed by team member (Yudan Xue), I executed
- **S3：均值回归策略：**基于团队成员（薛玉丹）提出的均值回归概念，我执行了

the complete technical conversion by writing s3_mean_reversion.py. This script implements a more complex statistical arbitrage logic: in __init__, it defines the 30-day moving average and standard deviation to calculate a Z-score. The next method then checks multiple conditions for entry: a trade is triggered only when the Z-score is below -2.0 (entry_threshold) **AND** the RSI is below 30 (rsi_oversold). The position will exit when the Z-score reverts to +0.5 (exit_threshold) or the holding period exceeds 20 days (max_holding_days). The core parameters are:

通过编写s3_mean_reversion.py实现完整的技术转换。该脚本采用更复杂的统计交易逻辑：在初始化阶段，系统会定义30日移动平均线和标准差以计算Z值。随后的交易策略需满足多重条件：只有当Z值低于-2.0（入场阈值）**且**RSI指标低于30（超卖状态）时才会触发交易。当Z值回升至+0.5（退出阈值）或持仓周期超过20天（最大持仓天数）时，系统将自动平仓。核心参数设置如下：

- **S4: CVaR Combined Strategy:** I architected and wrote the most complex component for Part 2, combined_cvar_strategy.py. Responding to the concept proposed by team member (Haw) to use **CVaR (Conditional Value at Risk)** for portfolio optimization, I further researched and proposed using the **SLSQP (Sequential Least Squares Programming) algorithm** as the specific technical implementation for this minimization problem. In the code, I implemented a CVaROptimizer module. This module gathers the last 60 days (lookback_days) of returns every 20 trading days (rebalance_frequency), then calls the scipy. optimize. minimize(method='SLSQP') function to solve for the portfolio weights that minimize CVaR, subject to the 10%-70% (min_weight, max_weight) allocation constraints. The core parameters are:
- **S4：CVaR组合策略：**我负责设计并编写了第二部分combined_cvar_strategy.py中最复杂的组件。针对团队成员（Haw）提出的使用**CVaR（条件风险价值）**进行投资组合优化的概念，我进一步研究并提出采用**SLSQP（顺序最小二乘规划）算法**作为该最小化问题的具体技术实现方案。在代码中，我实现了CVaROptimizer模块。该模块以每20个交易日为周期（rebalance_frequency）汇集过去60个交易日（lookback_days）的收益率数据，随后调用scipy. optimize.minimize（method=‘SLSQP’）函数，在10%-70%（min_weight，max_weight）分配约束条件下求解使CVaR最小化的投资组合权重。核心参数设置如下：

3. Analysis, Visualization & Report Preparation

3. 分析、可视化与报告编制

I provided the **complete writing framework** for the Design Report and was responsible for generating all data and visualizations contained within it. By writing the generate_strategy_analysis.py script, I utilized the pandas library for in-depth statistical analysis (e.g., calculating rolling means, Z-scores, RSI) and used matplotlib to automatically generate all core data analysis charts for Part 1 (Fig 1.1, 2.1, 3.1). Concurrently, I wrote the create_equity_comparison.py and create_comparison_chart.py scripts. These scripts extracted performance metrics (e.g., total return, Sharpe ratio, max drawdown) from the run_summary.json files and used matplotlib to plot all summary visualizations for Part 2 (Fig 7.1, 7.2), as well as comparison tables like all_strategies_performance_table.png not featured in the final report.

我提供了**设计报告的完整写作框架**，并负责生成其中包含的所有数据和可视化图表。通过编写generate_strategy_analysis.py脚本，我利用pandas库进行了深入的统计分析（例如计算滚动均值、Z分数、RSI），并使用 matplotlib 自动生成了第一部分（图1.1、2.1、3.1）的所有核心数据分析图表。同时，我还编写了create_equity_comparison.py和create_comparison_chart.py脚本。这些脚本从 run_summary.json 文件中提取了绩效指标（例如总回报率、夏普比率、最大回撤），并使用 matplotlib 绘制了第二部分（图7.1、7.2）的所有汇总可视化图表，以及未在最终报告中出现的比较表格，如all_strategies_performance_table.png。

In terms of report writing, I personally authored the Pseudocode sections for all strategies and the entirety of the Preliminary Testing and Optimization chapters. In these testing sections, I designed and implemented a rigorous **data-splitting methodology** (70% train / 30% test split, i.e., Days 1-700 / 701-1000) and employed **systematic Grid Search** to find robust parameters (e.g., S1's fast_period and slow_period; S3's entry_threshold and exit_threshold), ensuring strategy effectiveness on out-of-sample (OOS) data and avoiding overfitting. For other data-driven sections (e.g., Asset Selection Process, Data Analysis (S1, S3), Combined Strategy Performance Results), I was responsible for **providing all the charts and data** and assisting team members (Zhang Ao, Yudan Xue,Haw) with their analysis to support their respective drafting. The final refinement of the report, such as formatting and proofreading,was a collaborative team effort.

在报告撰写方面，我亲自撰写了所有策略的伪代码部分以及整个预测试与优化章节。在这些测试章节中，我设计并实施了严格的**数据划分方法**（70%训练集/30%测试集，即第1-700天/701-1000天），并采用**系统化的网格搜索法**来确定稳健参数（例如S1的fast_period和slow_period、S3的entry_threshold和exit_threshold），确保策略在样本外数据上的有效性并避免过拟合。对于其他数据驱动章节（如资产筛选流程、数据分析（S1、S3）、组合策略业绩结果），我负责**提供所有图表和数据**，并协助团队成员（张奥、薛玉丹、霍先生）进行分析以支持各自的起草工作。报告的最终润色工作，如格式调整和校对，是团队协作的成果。

4. Automation Framework & Testing

4. 自动化框架与测试

To ensure the reproducibility, consistency, and compliance of all backtests, I wrote and maintained the run_all_strategies.py script. Its implementation logic involved defining a configuration list of all strategy names and output paths, then iteratively calling subprocess.run() method to automate the batch execution of all four strategies, generating uniform performance reports like all_strategies_summary.md.

为确保所有回测的可重复性、一致性和合规性，我编写并维护了run_all_strategies.py脚本。其实现逻辑包括：定义所有策略名称和输出路径的配置列表，然后通过迭代调用subprocess.run（）方法来自动化批量执行所有四种策略，生成all_strategies_summary.md这类统一的性能报告。

ZHANG AO
ZHANG 原发性开放性青光眼

Report, Data, and Framework Contributions

报告、数据与框架贡献

At the beginning of the project, I actively participated in report writing, data analysis and framework research. I analysed the Data1 data set, examined all ten-asset series and visualised their price trends. This helped me understand the long-term directional pattern and fluctuation structure. Exploratory analysis not only clarifies the differences in the price trends of different assets but also guides our strategic allocation decision-making in the first part.

项目启动初期，我全程参与报告撰写、数据分析及框架研究工作。通过深入分析Data1数据集，系统梳理所有十类资产的系列数据，并绘制价格走势图，从而清晰把握长期趋势走向与波动特征。这种探索性分析不仅揭示了不同资产价格走势的差异性，更为前期的战略配置决策提供了重要依据。

Based on these findings, I drafted the strategy type definition of Strategy 1, and used the chart provided by Zhao Yixuan to write a data analysis section to describe the trend characteristics of Series 04 and its applicability to momentum trading based on the moving average. In addition, I have also optimised the format and presentation of Strategy 1 to ensure that it is clear and easy to read and consistent with the overall report structure.

基于这些研究发现，我制定了策略1的类型定义框架，并借助赵一轩提供的图表编写了数据分析章节，系统阐述了04系列指标的趋势特征及其在动量交易中的应用价值（基于移动平均线）。同时，我还优化了策略1的格式与呈现方式，确保其内容清晰易懂，与整体报告架构保持逻辑统一。

In the "Plan" section, I wrote a plan for strategy 1 and combination strategies, and in the second part, I outlined how to extend these strategies to portfolio optimisation based on CVaR. I also participated in the design presentation discussion and assisted in the production of presentation slides to show the analytical framework, core strategy logic and implementation process of

在‘计划’章节中，我制定了针对策略1及组合策略的实施方案；在第二部分，详细阐述了如何基于条件价值风险（CVaR）将这些策略拓展至投资组合优化领域。此外，我还参与了方案设计的讨论环节，并协助制作演示文稿，系统呈现了分析框架、核心策略逻辑及实施流程。

our project. In addition, I have also conducted literature research on trend tracking and risk adjustment optimisation methods to support our theoretical framework and ensure that our projects comply with established academic principles. Finally, I have proofread the whole report to ensure the technical accuracy, coherence and style consistency of each part.

此外，我还针对趋势追踪与风险调整优化方法开展了文献调研，以支撑我们的理论框架，并确保项目符合既定的学术规范。最后，我通读了整份报告，确保各部分的技术准确性、逻辑连贯性及风格统一性。

In addition to writing reports, I also conducted background research and framework analysis. I studied the budget limit and bankruptcy rules, order types and the BT396 backtest framework to ensure that it technically meets the requirements of COMP396. I also reviewed nine representative examples in the BT396 strategy and indicator library, and summarised the logic behind momentum, mean regression and spread strategies. This process helped me understand the core mechanisms of order execution, parameter calibration and risk control in algorithmic strategies. These tasks together lay the theoretical and technical foundation for the team's subsequent strategic development.

除了撰写报告，我还进行了背景调研和框架分析。我深入研究了预算限制、破产规则、订单类型以及BT396回测框架，确保其技术层面符合COMP396的要求。同时，我审阅了BT396策略与指标库中的九个典型案例，并总结了动量策略、均值回归策略和价差策略背后的逻辑原理。这一过程帮助我深入理解了算法策略中订单执行、参数校准和风险控制的核心机制。这些工作共同为团队后续的战略发展奠定了理论与技术基础。

Strategy Design and Technical Implementation

策略设计与技术实现

On this basis, I designed and tested the Dual Moving Average Crossover strategy for Series 04, which is the trend-following component of our overall trading framework. This method draws on the established momentum effect, where assets with strong performance in recent years tend to maintain their trend direction [1]. Our analysis revealed that Series 04 showed an alternate upward and downward trend, confirming that it was an ideal choice for moving-average-based trading systems.The strategy used a 3-day fast moving average(MA) and a 24-day slow moving average, generating golden and death cross signals for position entry and exit. I implemented a 10% stop-loss and 20% take-profit, forming a realistic 2:1 reward-to-risk ratio, and achieved an annualised Sharpe ratio of 2.013 with a maximum drawdown of 2.58% [11].

基于此，我为Series 04系列设计并测试了双移动平均线交叉策略，这是我们整体交易框架中的趋势跟踪组件。该方法借鉴了成熟的动量效应理论，即近年来表现强劲的资产往往能保持其趋势方向[1]。我们的分析显示，Series 04系列呈现出交替的上升与下降趋势，证实其是基于移动平均线交易系统的理想选择。该策略采用3日快速移动平均线和24日慢速移动平均线，通过生成黄金交叉和死亡交叉信号来确定入场与离场时机。我设置了10%的止损和20%的止盈，形成了实际的2：1风险回报比，并实现了年化夏普比率2.013，最大回撤率2.58%[11]。

To ensure robustness and prevent overfitting, I conducted systematic parameter optimization through grid search with a 70/30 training-test split, prioritizing parameter stability over backtest profit maximization. The evaluation incorporated multiple performance metrics—including Sharpe ratio, maximum drawdown, and win rate—following established best practices [7]. Additional sensitivity analyses across varying lookback periods and trade durations verified the model's consistent profitability. After finalizing the configuration, I delivered the findings and a streamlined implementation plan to the development team for integration into the BT396 framework, ensuring full compliance with our shared risk management standards.

为确保模型稳健性并防止过拟合，我采用70/30的训练-测试分割比例，通过网格搜索进行系统性参数优化，优先考虑参数稳定性而非回测收益最大化。评估指标包含夏普比率、最大回撤和胜率等多项标准，严格遵循行业最佳实践[7]。通过不同回溯周期和交易时长的敏感性分析，进一步验证了模型的持续盈利能力。最终确定配置方案后，我向开发团队提交了研究成果及优化后的实施方案，确保其能无缝集成至BT396框架，并完全符合我们共同制定的风险管理规范。

During design presentation sessions, I articulated the conceptual framework of the crossover system, presented comprehensive parameter testing results, and proposed integration pathways for the Part 2 Combined Strategy to incorporate CVaR-based portfolio optimization—demonstrating how this approach would enhance tail-risk management capabilities across our trading framework [10].

在设计演示环节中，我系统阐述了交叉系统的核心架构，展示了全面的参数测试结果，并为第二部分的综合策略提出了整合方案，将基于条件价值风险（CVaR）的投资组合优化纳入其中——通过具体案例说明，这种创新方法将显著提升我们交易框架中的尾部风险管理能力[10]。

Reflection and Learning

反思与学习

This project enables me to combine theoretical knowledge with empirical practice. It deepens my understanding of how momentum models, parameter robustness and pullback control interact in real trading environments, and how the backtest framework transforms these concepts into quantitative results. In addition to technological progress, I have also improved my ability to manage quantitative research tasks, systematically conduct parameter verification, and effectively communicate research results in the team.

这个项目让我实现了理论与实践的完美结合。它不仅让我深入理解了动量模型、参数稳健性与回撤控制在真实交易环境中的协同作用，更让我掌握了如何将这些概念转化为量化结果的回测框架。除了技术能力的提升，我还系统掌握了量化研究任务的管理方法，能够规范开展参数验证工作，并在团队中高效传达研究成果。

Through collaborative discussion and report writing, I learned how to transform analytical insights into clear and rigorous text results to ensure that the technical content is easy to understand for readers with quantitative analysis ability and non-technical background. In general, this experience has enhanced my quantitative reasoning ability, academic writing ability and interdisciplinary coordination ability, and also improved my ability to integrate personal research results into the data-oriented overall project portfolio design process.

通过协作讨论与报告撰写实践，我掌握了将分析洞见转化为清晰严谨文本成果的技巧，确保技术内容能被具备定量分析能力的读者及非技术背景读者轻松理解。总体而言，这段经历不仅提升了我的定量推理能力、学术写作水平与跨学科协调能力，更增强了将个人研究成果融入数据导向型整体项目组合设计流程的能力。

WONG HAW
黄花

I served as the team's Lead Strategy Designer and Analyst, responsible for the foundational research, conceptual design, and risk management framework of the project. My contributions focused on identifying viable trading paradigms through data analysis, researching advanced portfolio construction methodologies, and authoring the strategic justifications and future development plans for the team's key strategies.

我担任团队首席战略设计师兼分析师，负责项目的基础研究、概念设计及风险管理框架构建。主要工作包括：通过数据分析识别可行的交易模式、研究先进的投资组合构建方法，并为团队核心战略撰写战略论证及未来发展规划。

In the initial phase, background research focused on the project's core failure points, specifically the high rate of out-of-sample performance decay noted in Lecture 7 and the bankruptcy risks demonstrated in the ‘p_bankrupt.py’ example. Furthermore, our preliminary visual analysis exhibited the price patterns of series 05 had a strong, persistent positive drift over the long term. This led me to propose the "Buy & Hold" (B&H) strategy as one of our individual strategies and this B&H analysis served as the essential quantitative baseline against which all active strategies were measured.

在项目初期，我们通过背景调研锁定核心风险点：第七讲中提到的样本外表现衰减率居高不下，以及“p_bankrupt.py”案例揭示的破产风险。通过初步可视化分析发现，05系列的价格走势长期呈现强劲且持续的正向偏移。基于这些发现，我提出将“买入并持有”（B&H）策略作为我们的核心策略之一，该策略的量化分析结果被确立为基准线，所有主动策略的表现都将以此为参照进行评估。

My primary contribution was the conceptual design of the "Combined Strategy." I identified that simply aggregating the team's individual strategies would expose the portfolio to significant correlation risk, which would be detrimental to the project's key performance metric, the PD Ratio. To solve this, I have conducted a comprehensive comparative analysis of portfolio construction methodologies, evaluating the pros and cons of Mean-Variance Optimization (MVO) and Hierarchical Risk Parity (HRP) models. For MVO, this well-established classic Markowitz model was unsuitable for us because it relied on variance as the sole measure of risk

我的主要贡献是“综合策略”的概念设计。我认识到，简单地汇总团队的个人策略会使投资组合面临显著的相关性风险，这将对项目的业绩指标——PD比率产生不利影响。为了解决这个问题，我进行了全面的投资组合构建方法比较分析，评估了均值-方差优化（MVO）和层次风险平价（HRP）模型的优缺点。对于MVO，这个成熟的经典马科维茨模型并不适合我们，因为它仅依赖方差作为风险的唯一衡量标准。

treats upside "good" volatility and downside "bad" volatility as a-historical, and it is notoriously unstable to small changes in input assumptions[9]. On the other hand, for HRP, this is a modern, machine-learning-based method that first clusters correlated strategies. This method will cluster assets by similarity (e.g., equities together, bonds together), form a tree, allocate risk first across major clusters, then split within clusters. This will prevent overweighting many highly correlated assets. While academically interesting, I determined it was likely too complex to implement and justify for a portfolio of just three strategies[8].

将上行“好”波动性和下行“坏”波动性视为非历史性的，并且对输入假设的小幅变化极其不稳定[9]。另一方面，对于HRP而言，这是一种现代的、基于机器学习的方法，首先将相关策略聚类。该方法会根据相似性对资产进行聚类（例如，将股票归为一类，将债券归为一类），形成树状结构，首先在主要聚类间分配风险，然后在聚类内进行细分。这将防止过度配置许多高度相关的资产。尽管在学术上很有趣，但我认为对于只有三种策略的投资组合来说，这种方法可能过于复杂，难以实施和证明其合理性[8]。

Based on this research, I formally proposed the adoption of Conditional Value-at-Risk (CVaR) to our code developer as the team's core risk management methodology. My justification was that CvaR, as a risk measure, is superior to variance because it specifically quantifies the expected loss in the worst-case tail scenarios. And CVaR focus on minimizing tail risk directly aligns with the project's objective of minimizing maximum drawdown. I have designed complete conceptual architecture for this strategy, with the help from our Lead Developer I, specifying how the three individual strategies would serve as inputs to a CVaR optimization layer.

基于这项研究，我正式向代码开发团队提出采用条件风险价值（CVaR）作为核心风险管理方法。我的理由是：作为风险度量工具，CVaR比方差更具优势，因为它能精准量化最坏情况下的尾部场景预期损失。而CVaR专注于最小化尾部风险，这与项目“最小化最大回撤”的目标直接契合。在首席开发人员I的协助下，我为该策略设计了完整的概念架构，明确了三种独立策略如何作为输入数据为CVaR优化层提供支持。

I was also responsible for authoring the "Future Plan" for Data 2, demonstrating strategic foresight. I proposed evolving static models into adaptive systems for both individual strategies and combined strategy. For the individual strategies, I designed refinement hypotheses to be tested with Data 2, including mitigating the B&H strategy's "no risk management" weakness by introducing a 200-day moving average as a macro regime filter. Furthermore, I suggested to Zhang Ao and Xue Yudan that using an ADX (Average Directional Index) filter to solve the "whipsaw" conflicts in the Mean-Reversion and Trend-Following strategies, allowing them to trade only in appropriate market regimes. In addition, for the Combined Strategy, I identified its vulnerability to overnight price gaps and proposed researching the integration of Limit Orders to manage slippage.

我还负责为数据2撰写“未来计划”，展示了战略远见。我提议将静态模型演变为适应性系统，适用于单一策略和组合策略。对于单一策略，我设计了改进假设以在数据2中测试，包括通过引入200日移动平均线作为宏观制度过滤器来缓解B&H策略的“无风险管理”弱点。此外，我建议张奥和薛玉丹使用ADX（平均方向指数）过滤器来解决均值回归和趋势跟踪策略中的“鞭锯”冲突，使他们只能在适当的市场制度下交易。另外，对于组合策略，我指出了其对隔夜价格缺口的脆弱性，并提议研究整合限价单以管理滑点。

Finally, in preparing the Design Presentation Report, I was the primary author for the "Strategy 2: Buy & Hold" and helping out our Lead Developer, I, to finish the "Combined Strategy" sections together. For the CVaR portion, I wrote the complete theoretical justification and the rationale for its selection. However, for the necessary technical implementation (SLSQP) which was handled by the Lead Developer I.

最后，在准备设计演示报告时，我是“策略2：买入并持有”的主要作者，并协助我们的首席开发人员I. 完成“综合策略”部分。对于CVaR部分，我撰写了完整的理论依据及其选择的理由。然而，对于必要的技术实现（SLSQP），由首席开发人员I. 负责。

XUE YUDAN
血 YUDAN

Guided by the CANVAS Rules & Policies—Budget Limit and Bankruptcy, Order Types, Plagiarism and Collusion—the strategy is framed and evaluated under these constraints. From the instructor’s starter strategies and rule library in COMP396, I drew three key ideas that led me to choose mean-reversion: from bbands contrarian, I adopted the idea that “outside the band = extreme deviation,” then used Z-score and RSI as confirmations (inspired by Saria’s idea); from bbands holding period diag, I learned to take profits when price re-enters the band or touches the middle line, and to set a maximum holding period to avoid getting stuck; and from fixed and bankrupt, I took the discipline of position caps, budget limits, and protection rules—using bbands trend following as a contrast that reminds me to cut or pause contrarian trades during clear trend phases. Combining these insights with all charts Saria generated across the datasets gave me confidence that mean-reversion is a workable framework here.

在《Canvas规则与策略——预算限制与破产、订单类型、抄袭与合谋》的框架指引下，我们构建并评估了这套策略体系。从COMP396课程中导师提供的入门策略和规则库中，我提炼出三个核心理念促使我选择均值回归策略：从BBands反向交易策略中，我采纳了“超出区间即为极端偏离”的理念，并运用Z分数和RSI指标进行验证（灵感源自萨里亚的思路）；从BBands持仓周期分析中，我领悟到当价格重新进入区间或触及中线时及时获利了结，并设置最大持仓周期以避免陷入困境；从固定止损与破产机制中，我借鉴了仓位上限、预算限制和保护规则的纪律性——通过对比BBands趋势跟踪策略，提醒自己在趋势清晰阶段及时止损或暂停反向交易。将这些洞见与萨里亚在数据集中生成的所有图表相结合，使我确信均值回归策略在此框架下具有可行性。

By reading Trading Systems, 2nd ed., I learned from the Equity Curve MA Switch (Ch. 8.3–8.4) to pause the strategy when the equity curve falls below its moving average and resume when it rises back above—check the environment first. Based on that, the Walk-Forward Activator (Ch. 6.2–6. 3) shows the need for rolling re-estimation of parameters to help reduce overfitting[5]. Then, Quantitative Trading (CN) §2.8 on statistical arbitrage/pairs trading provides evidence that deviations tend to revert, while stressing strict stop-losses and exiting during adverse phases [13]. Following these three points, I conclude that a mean-reversion framework is a good fit for this project.

通过研读《交易系统》第二版，我从股票曲线移动平均线切换策略（第8.3-8.4章）中领悟到：当股票曲线跌破移动平均线时暂停策略，待其回升至均线之上再继续执行——需先评估市场环境。基于此，前向步进激活器（第6.2-6.3章）揭示了参数滚动重估的必要性，以帮助减少过拟合[5]。随后，《量化交易（CN）》第2.8节关于统计套利/货币对交易的论述，不仅提供了偏差会回归的实证依据，更强调在不利阶段必须严格执行止损并及时退出[13]。综合这三点，我认为均值回归框架非常适合本项目应用。

Strategy Proposal

战略建议

From our team’s charts, I find that the series 09 often “bounces back” whenever price touches the upper or lower band, and overall it moves up and down around its moving average. Based on this, and after reading Trading Systems (2nd ed.), I suggest using a mean-reversion strategy as Strategy 3 and write the strategy type definition of it.

根据团队技术分析图表，我们发现09系列指标在价格触及上下边界时往往会出现反弹，整体走势围绕移动平均线上下波动。基于此观察，并参考《交易系统》（第二版）的分析，建议采用均值回归策略（即策略3），并制定该策略的类型定义。

Data Analysis

数据分析

Looking at Series 09 mean-reversion indicators with Bollinger Bands and Z-score, plus the Z-score distribution and “Z-score vs Future Returns” plots, I find: Most of the time, price moves around the moving average and often comes back after touching the upper or lower band. And when Z-score hits extreme levels, it usually comes back toward 0. In a few periods, price “rides the band” in one direction. In those times, mean-reversion is weaker. The regression line in the scatter plot slopes slightly downward, suggesting: the more extreme the move, the more likely a partial pullback—but the effect is not extraordinarily strong. So, I can confirm Series 09 fits mean-reversion and write the data analysis part of the strategy 3.

通过分析09系列数据的均值回归指标（结合布林带和Z值）、Z值分布图以及“Z值与未来收益”对比图，我发现：多数情况下价格会在移动平均线附近波动，触碰布林带上下轨后往往会回调。当Z值达到极端水平时，通常会回归至0附近。在少数时段，价格会单向“横冲直撞”于布林带内，此时均值回归效应较弱。散点图中的回归线略微向下倾斜，表明：价格波动越剧烈，出现部分回调的可能性越大——但这种影响并不特别显著。因此，我确认09系列数据符合均值回归规律，并将数据分析部分纳入策略3。

Strategy Design

战略设计

Drawing on the observed chart patterns, I propose applying the strategy as follows: enter only when the Z-score reaches an extreme, using RSI as confirmation to avoid jumping in too early; take profits once price re-enters the band or touches the middle line—there’s no need to wait for a perfect return to the exact average; if price starts riding the band in a one-way move, treat it as a trend filter and reduce size or pause trading; and manage risk by setting a clear stop-loss and a maximum holding period to handle cases where the expected reversion doesn’t materialize.

基于观察到的图表形态，我建议采用以下策略：仅在Z值达到极端值时入场，并以相对强弱指数（RSI）作为确认信号，避免过早入场；当价格重新进入波段或触及中线时即可获利了结——无需等待价格完全回归均值；若价格开始单向突破波段，应将其视为趋势过滤器并缩减仓位或暂停交易；同时通过设置明确止损位和最长持仓期限来管理风险，以应对预期回归未实现的情况。

Plan

计划

I recognize the unreleased data is unpredictable, so we need a regime indicator to decide when mean-reversion should be active. Based on the literature, I write the plan about the future plan of strategy 3 we used, and propose to use the Average Directional Index (ADX) as a filter: mean-reversion trades only when ADX is low (range-bound market) and stands aside or reduces size when ADX is high (strong trend) [13]. The entry rule stays the same—Z hits an extreme and RSI confirms—while the exit remains "price re-enters the band or touches the middle line".

我认识到未发布的数据具有不可预测性，因此我们需要一个制度指标来决定均值回归何时应该启动。根据文献资料，我撰写了关于未来策略3的计划，并提议使用平均方向指数（ADX）作为过滤器：当 ADX 较低时（区间市场），均值回归交易；当 ADX 较高时（强趋势），均值回归则保持观望或减小规模[13]。入场规则保持不变——Z触及极端且RSI确认——而出场条件仍为"价格重新进入区间或触及中线"。

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